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  CodeAmong — Bundle đầy đủ tất cả file Logic & UI
  Nguồn: GgUI_v6-2.html (đã tách) + tính năng Meeting Vote & Taskbar
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┌─                                FILE                                TREE
──────────────────────────────────────────────────────────────────────────────
──────────────┐
│  codreamong/
│   ├── index.html                  [CONFIG]   Entry point — load scripts theo thứ tự
│   ├── style.css                   [CSS]      Toàn bộ visual styling
│   ├── README.md                              Hướng dẫn kiến trúc
│   │
│   └── js/
│       ├── constants.js            [LOGIC]    Dữ liệu tĩnh: COLORS, FILES, PUBLIC_ROOMS
│       ├── state.js                [LOGIC]    Object S — nguồn sự thật duy nhất
│       ├── network.js              [LOGIC]    PeerJS WebRTC transport
│       ├── game.js                 [LOGIC]    Phase transitions, timers, role assignment
│       ├── tasks.js                [LOGIC]★   Task system: định nghĩa + logic hoàn thành
│       ├── meeting-vote.js         [LOGIC]★   Vote cuộc họp + network messages
│       ├── actions.js              [LOGIC]    Controller — bridge giữa DOM và logic
│       │
│       └── ui/
│           ├── render.js           [UI]       render() dispatcher + global overlays
│           ├── lobby.js            [UI]       Lobby phase + create/join modals
│           ├── waiting.js          [UI]       Waiting room + chat + settings
│           ├── role.js             [UI]       Role reveal: Dev vs VibeCoder screen
│           ├── taskbar.js          [UI] ★     Task sidebar + progress strip
│           ├── meeting-vote-ui.js[UI] ★      Prompt nhỏ + banner vote họp
│           └── coding.js           [UI]       Coding phase: editor + overlays
│
│  ★ = File mới thêm trong lần cập nhật này
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┌─                      LUỒNG                      PHỤ                    THUỘC
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┐
│                                                                   constants.js
──────────────────────────────────────────────────────────────────────────┐
│                                                 state.js                  (S{})
──────────────────────────────────────────────────────────────────────────│
│       ↓ đọc/ghi bởi:                                                     │
│       network.js → handleNetworkMessage() → game.js / meeting-vote.js   │
│       game.js    → enterCodingPhase() → tasks.js.initTasks()            │
│       actions.js → doOpenMeeting() → meeting-vote.js.requestMeeting()  │
│                                                                         ↓
│  ui/render.js.render()                                          (reads all)
│       ├── ui/lobby.js.renderLobby()
│       ├── ui/waiting.js.renderWaiting()
│       ├── ui/role.js.renderRolePhase()
│       └── ui/coding.js.renderCoding()
│               ├── ui/taskbar.js.renderTaskbar() + renderProgressStrip()
│                                   └── ui/meeting-vote-ui.js.buildMeetingPrompt()  +
buildMeetingVoteBanner()
```

Chỉnh sửa bằng WPS Office

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[CONFIG] [ENTRY POINT]
# index.html

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► EXPORT / KHAI BÁO:
— (không export hàm, chỉ khai báo thứ tự load script)

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
window.onload → render(), initPeer()

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[ BẮT ĐẦU: index.html ]

```
<!DOCTYPE html>
<html lang="vi">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CodeAmong - VS Code Game</title>

    <!-- External dependencies -->
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://unpkg.com/peerjs@1.4.7/dist/peerjs.min.js"></script>

    <!-- Styles -->
    <link rel="stylesheet" href="style.css">
</head>
<body>

<!-- Single mount point. All UI is rendered into here by render.js -->
<div id="app" style="display:flex;flex-direction:column;height:100vh;"></div>

<!--
    Script load order matters — each file depends on the ones above it:

    1. constants.js    — pure data (COLORS, FILES, PUBLIC_ROOMS)
    2. state.js        — global S object (reads COLORS)
    3. network.js      — P2P transport (reads/writes S, calls game.js fns)
    4. game.js         — game logic & timers (reads/writes S, calls render)
    5. actions.js      — UI event handlers (calls game.js + network.js)
    6. ui/render.js    — main render() dispatcher
    7. ui/lobby.js     — lobby phase HTML
    8. ui/waiting.js   — waiting room HTML
    9. ui/role.js      — role reveal HTML
   10. ui/coding.js    — coding phase HTML

    ── Separation of concerns ──────────────────────────────────────
    UI files (ui/*.js)  → only return HTML strings. Zero side effects.
    game.js             → owns timers and phase transitions.
    network.js          → owns all P2P I/O.
```

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```
    actions.js              → thin glue between DOM events and game/network.
    state.js                → single source of truth; never import — global S.

    ── What would move to a real backend ──────────────────────────────
    network.js      → replaced by WebSocket client (socket.io etc.)
    game.js         → server owns: role assignment, timer, vote aggregation
    The client would only: send user actions → receive state → render()

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```

[ BẮT ĐẦU: style.css ]

```
/* ============================================================
   CodeAmong — style.css
   All visual styling. Zero game logic here.
   ============================================================ */

@import
url('https://fonts.googleapis.com/css2?family=JetBrains+Mono:wght@300;400;500;600;700&
display=swap');

* {
    font-family: 'JetBrains Mono', monospace;
    box-sizing: border-box;
}

body {
    background: #1e1e1e;
    color: #cccccc;
    height: 100vh;
    overflow: hidden;
    display: flex;
    flex-direction: column;
    margin: 0;
}

/* ── Animations ── */
.modal-anim {
    animation: modalPop 0.18s cubic-bezier(.4, 0, .2, 1);
}
@keyframes modalPop {
    from { transform: scale(0.93) translateY(8px); opacity: 0; }
    to   { transform: scale(1) translateY(0); opacity: 1; }
}

.countdown-anim {
    animation: countPop 0.4s cubic-bezier(.4, 0, .2, 1);
}
@keyframes countPop {
    0%   { transform: scale(2.5); opacity: 0; }
    60%  { opacity: 1; }
    100% { transform: scale(1); opacity: 1; }
}

@keyframes vignettePulse {
    from { opacity: 0.6; }
    to   { opacity: 1; }
}
@keyframes imposterReveal {
    0%   { transform: scale(0.6); opacity: 0; }
    60%  { transform: scale(1.08); }
    100% { transform: scale(1); opacity: 1; }
}
@keyframes blink {
    0%, 100% { opacity: 1; }
```

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```
    50%         { opacity: 0.3; }
}
@keyframes speakPulse {
    from { box-shadow: 0 0 0 2px #22c55e40; }
    to   { box-shadow: 0 0 0 8px #22c55e80; }
}

/* ── Form elements ── */
textarea {
    font-family: 'JetBrains Mono', monospace;
    font-size: 14px;
    line-height: 1.6;
    tab-size: 4;
}

input[type=text],
input[type=number] {
    background: #2d2d2e;
    border: 1px solid #3d3d3d;
    color: #ccc;
    border-radius: 8px;
    padding: 8px 12px;
    outline: none;
    width: 100%;
    font-family: 'JetBrains Mono', monospace;
    font-size: 13px;
}
input:focus { border-color: #007acc; }

::-webkit-scrollbar { width: 5px; height: 5px; }
::-webkit-scrollbar-thumb { background: #444; border-radius: 3px; }

/* ── Buttons ── */
.btn {
    cursor: pointer;
    border: none;
    font-family: 'JetBrains Mono', monospace;
    font-size: 12px;
    border-radius: 8px;
    padding: 8px 16px;
    transition: all 0.12s;
}
.btn:active { transform: scale(0.96); }

.btn-blue     { background: #007acc; color: white; }
.btn-blue:hover     { background: #0090e8; }

.btn-green    { background: #22c55e; color: #000; }
.btn-green:hover    { background: #16a34a; color: #fff; }

.btn-red      { background: #ef4444; color: white; }
.btn-red:hover      { background: #dc2626; }

.btn-gray     { background: #37373d; color: #ccc; }
.btn-gray:hover     { background: #484848; }

.btn-purple { background: #a855f7; color: white; }
```

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```
.btn-purple:hover { background: #9333ea; }

.btn-amber    { background: #f59e0b; color: #000; }
.btn-amber:hover    { background: #d97706; }

/* ── Layout helpers ── */
.player-row {
    display: flex;
    align-items: center;
    gap: 12px;
    background: #1e1e1e;
    border: 1px solid #2d2d2e;
    border-radius: 14px;
    padding: 10px 14px;
}

.vscode-item {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 10px 16px;
    cursor: pointer;
    color: #ccc;
    transition: background 0.1s;
}
.vscode-item:hover { background: #2d2d2e; color: white; }

.fit-text { overflow: hidden; white-space: nowrap; }
.fit-sm     { font-size: 10px !important; }
.fit-md     { font-size: 12px !important; }

.tab-active {
    border-bottom: 2px solid #007acc;
    background: #1e1e1e;
    color: white;
}

/* ── Peer / network status ── */
.peer-dot {
    width: 8px;
    height: 8px;
    border-radius: 50%;
    display: inline-block;
}
.peer-ok     { background: #22c55e; box-shadow: 0 0 6px #22c55e; }
.peer-wait { background: #f59e0b; animation: blink 1s infinite; }
.peer-err    { background: #ef4444; }

/* ── Voice / speaking ── */
.speaking-ring { animation: speakPulse 0.6s infinite alternate; }

/* ── Utility ── */
.copy-btn {
    cursor: pointer;
    font-size: 10px;
    color: #007acc;
    padding: 2px 8px;
```

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```
    border: 1px solid #007acc;
    border-radius: 6px;
}
.copy-btn:hover { background: #007acc; color: white; }
```

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```
    },
    'game.cpp': {
        content: `#include<iostream>\nusing namespace std;\nint main(){\n
cout<<"C++"<<endl;\n}`,
        lang: 'cpp',
        icon: '⚙'
    }
};

// Public room list.
// In a real backend this comes from a lobby/matchmaking API.
const PUBLIC_ROOMS = [
    { id: '123456', name: 'Battle Royale #001', cur: 3, max: 5 },
    { id: '789012', name: 'Code Duel #042',     cur: 1, max: 5 },
    { id: '345678', name: 'Speed Code VIP',     cur: 4, max: 5 },
];
```

[ KẾT THÚC: js/constants.js ]

[LOGIC] [STATE]
# js/state.js

► EXPORT / KHAI BÁO:
S{} — object toàn cục chứa toàn bộ state

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
Được đọc/ghi bởi tất cả file logic và UI

[ BẮT ĐẦU: js/state.js ]

```
/* ============================================================
   state.js — single source of truth
   All mutable state lives here.
   UI reads from S. Game logic writes to S. Network syncs S.
   ============================================================ */

const S = {
    // ── Network (managed by network.js) ──────────────────
    peer:        null,          // PeerJS Peer instance
    peerId:      null,          // our assigned peer ID (string)
    peerStatus:  'connecting', // 'connecting' | 'ready' | 'error'
    connections: {},            // { peerId: DataConnection }

    // ── Local player identity ────────────────────────────
    myName:   'Player',
    myColor:  COLORS[0],
    myAvatar: AVATARS[0],

    // ── Room / session (synced by host via P2P) ──────────
```

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```
phase:          'lobby',        // 'lobby' | 'waiting' | 'role' | 'coding'
isHost:         false,
roomId:         null,
roomName:       '',
maxPlayers: 5,
players:        [],             // [{ peerId, name, color, avatar, isHost, readyVote }]

// ── UI sub-states (local only, not synced) ──────────────────────
modal:                  null,     // null | 'create' | 'join' | 'confirm-exit' | 'settings'
showChat:               false,
showMeetingChat: false,
voiceActive:            false,

// ── Chat messages (append-only, synced via CHAT messages) ─
chatMessages:           [],
meetingChatMessages: [{ from: 'System', msg: 'Meeting started!', color: '#858585' }],

// ── Coding editor (local) ──────────────────────────────────────────
currentFile:            'main.game.js',
tabsMinimized:          false,
showTerminal:           false,
terminalOutput:         '',
showMeeting:            false,
selectedVote:           null,
fileActivePlayers: {},     // { filename: [{ peerId, name, color }] }

// ── Game settings (host sets, synced to members) ──────────────
roundTime:      60,
meetingTime: 30,

// ── Coding timer (replicated on all clients) ──────────────────
codingTimer:            300, // seconds remaining
codingTimerRunning: false,
showRunResult:          false,
runResultSecs:          15,
voteSkipCounts:         0,

// ── Role phase ────────────────────────────────────────────────────
myRole:                 null,     // 'dev' | 'vibecoder'
vibeCoderPeerId: null,     // exactly one vibe coder per game
roleTimer:              null,     // seconds left on role reveal screen

// ── Misc UI ──────────────────────────────────────────────────────
notification: null,         // { msg, isErr }
countdown:      null,           // 3 → 2 → 1 → null
debugLog:       [],             // array of strings for the debug panel

// ── Meeting vote state ────────────────────────────────────────────
meetingVoteActive:   false,   // is a vote in progress (shown to all)
meetingVoteYes:         0,
meetingVoteNo:          0,
myMeetingVote:          null,     // null | 'yes' | 'no'
showMeetingPrompt:   false,   // show private "Bạn có muốn họp?" box

// ── Task system ───────────────────────────────────────────────────
tasks:                  {},           // { filename: [{id, label, done}] } — populated by initTasks()
```

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```
    completedTaskIds: new Set(),
    taskProgress:       0,              // 0-100, only Dev tasks count
    totalDevTasks:      0,

    // ── Internal timer handles (never render these) ──────────────
    _countdownTimer:          null,
    _roleTimer:                null,
    _codingTimerInterval: null,
    _runResultTimer:          null,
};
```

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```
function initPeer() {
    dbg('Initializing PeerJS...');
    updateNetworkStatus('connecting');

    const peer = new Peer(genId(), {
        host:     '0.peerjs.com',
        port:     443,
        path:     '/',
        secure: true,
        debug:    1,
        config: {
            iceServers: [
                { urls: 'stun:stun.l.google.com:19302' },
                { urls: 'stun:stun1.l.google.com:19302' },
                { urls: 'stun:stun2.l.google.com:19302' },
                // TURN relay — needed for cross-network connections
                {  urls:  'turn:openrelay.metered.ca:80',                    username:
'openrelayproject', credential: 'openrelayproject' },
                {  urls:  'turn:openrelay.metered.ca:443',                   username:
'openrelayproject', credential: 'openrelayproject' },
                {       urls:       'turn:openrelay.metered.ca:443?transport=tcp',       username:
'openrelayproject', credential: 'openrelayproject' },
            ]
        }
    });

    S.peer = peer;

    peer.on('open', id => {
        S.peerId = id;
        updateNetworkStatus('ready');
        dbg('✅ Peer ready. ID = ' + id);
        render();
    });

    peer.on('connection', conn => {
        dbg('📥 Incoming connection from: ' + conn.peer);
        setupConn(conn);
    });

    peer.on('error', err => {
        dbg('❌ Peer error: ' + err.type + ' — ' + err.message);
        if (err.type === 'peer-unavailable') {
            notify('❌ Không tìm thấy phòng. Kiểm tra lại Room ID.', true);
        } else if (err.type === 'network' || err.type === 'server-error') {
            notify('⚠ Mất kết nối. Thử reload trang.', true);
            updateNetworkStatus('error');
        } else {
            notify('⚠ ' + err.type, true);
        }
        render();
    });

    peer.on('disconnected', () => {
        dbg('⚠ Disconnected. Reconnecting...');
        updateNetworkStatus('connecting');
```

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```
            setTimeout(() => { if (!S.peer.destroyed) S.peer.reconnect(); }, 2000);
        });
}

function updateNetworkStatus(status) {
        S.peerStatus = status;
        // Patch status bar in-place if it exists, otherwise full render picks it up
        const el = document.getElementById('peer-status-bar');
        if (el) renderStatusBarInto(el);
}

// ── Connection lifecycle ──────────────────────────────────────────────────

function setupConn(conn) {
        // Register data/close events immediately (before 'open') to avoid missing events
        conn.on('data', data => {
                dbg('📨 Recv: ' + data.type + ' from ' + conn.peer);
                handleNetworkMessage(data, conn.peer);
        });

        conn.on('close', () => {
                dbg('🔌 Closed: ' + conn.peer);
                delete S.connections[conn.peer];
                // Remove player from list
                S.players = S.players.filter(p => p.peerId !== conn.peer);
                // If we are host, tell everyone about the updated list
                if (S.isHost) sendToAll({ type: 'PLAYERS_UPDATE', players: S.players });
                addChatMessage('🔴 Một người chơi đã rời phòng', 'system');
                render();
        });

        conn.on('error', e => dbg('Conn error: ' + e));

        // Store as soon as open
        const onOpen = () => {
                dbg('🔗 Open: ' + conn.peer);
                S.connections[conn.peer] = conn;
        };
        if (conn.open) onOpen();
        else conn.on('open', onOpen);
}

function connectToPeer(targetId, onOpen) {
        dbg('🔌 Connecting to: ' + targetId);
        if (!S.peer || S.peer.destroyed) { notify('❌ Peer chưa sẵn sàng!', true); return; }
        if (S.peerStatus !== 'ready')       { notify('❌ Chờ kết nối mạng xong đã!', true); return; }

        const conn = S.peer.connect(targetId, { reliable: true, serialization: 'json' });
        setupConn(conn);
        conn.on('open',    () => onOpen(conn));
        conn.on('error', e    => { dbg('Connect error: ' + e); notify('❌ Không thể kết nối: ' + e, true);
 });

        setTimeout(() => {
                if (!conn.open) {
                        notify('❌ Timeout. Phòng không phản hồi.', true);
```

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```
            dbg('⏱ Timeout connecting to ' + targetId);
        }
    }, 8000);
}

// ── Send helpers ──────────────────────────────────────────────────────────

function sendTo(peerId, data) {
    const c = S.connections[peerId];
    if (c && c.open) c.send(data);
    else dbg('⚠ Cannot send to ' + peerId + ': not open');
}

function sendToAll(data) {
    Object.keys(S.connections).forEach(id => sendTo(id, data));
}

// ── Incoming message router ───────────────────────────────────────────────
//
//   Each case maps a network message type to a game-logic call.
//   UI is never touched here directly.
//
//   [BACKEND NOTE] In a server-based architecture:
//     - The server receives JOIN_REQUEST and decides who gets what role.
//     - Clients only send "intents" (join, vote, chat) and receive "state".
//     - The switch below would shrink to: update S from server payload → render().

function handleNetworkMessage(data, fromPeer) {
    switch (data.type) {

        // ── Room joining ──────────────────────────────────────────────────
        case 'JOIN_REQUEST': {
            // [BACKEND] Server would validate & persist this
            if (!S.isHost) return;
            if (S.players.length >= S.maxPlayers) {
                sendTo(fromPeer, { type: 'JOIN_DENIED', reason: 'Phòng đầy rồi!' });
                return;
            }
            const newPlayer = {
                peerId: fromPeer,
                name:     data.name,
                color:    data.color,
                avatar: data.avatar,
                isHost: false,
                readyVote: false
            };
            S.players.push(newPlayer);
            sendTo(fromPeer, {
                type: 'JOIN_ACCEPTED',
                players:       S.players,
                roomName:      S.roomName,
                maxPlayers: S.maxPlayers,
                roomId:        S.roomId
            });
            sendToAll({ type: 'PLAYERS_UPDATE', players: S.players });
            addChatMessage('🟢 ' + data.name + ' đã vào phòng', 'system');
            render();
```

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```
        break;
    }

    case 'JOIN_ACCEPTED': {
        S.players       = data.players;
        S.roomName      = data.roomName;
        S.maxPlayers = data.maxPlayers;
        S.roomId        = data.roomId;
        S.phase         = 'waiting';
        notify('✅ Đã vào phòng: ' + data.roomName);
        render();
        break;
    }

    case 'JOIN_DENIED': {
        notify('❌ Từ chối: ' + data.reason, true);
        break;
    }

    // ── Room state sync ──────────────────────────────────────────────
    case 'PLAYERS_UPDATE': {
        S.players = data.players;
        render();
        break;
    }

    case 'SETTINGS_UPDATE': {
        // [BACKEND] Server would validate and store round config
        S.roundTime      = data.roundTime;
        S.meetingTime = data.meetingTime;
        addChatMessage('⚙ Host cập nhật cài đặt game', 'system');
        render();
        break;
    }

    // ── Game flow ────────────────────────────────────────────────────
    case 'START_COUNTDOWN': {
        // [BACKEND] Server triggers this after all-ready check
        startCountdown();
        break;
    }

    case 'ROLE_ASSIGN': {
        // [BACKEND] Server picks vibe coder server-side (prevent cheating)
        // Here host acts as authoritative source instead
        applyRoles(data.vibeCoderPeerId);
        break;
    }

    // ── Voting ───────────────────────────────────────────────────────
    case 'VOTE_REQUEST': {
        // [BACKEND] Server aggregates votes and triggers countdown
        if (!S.isHost) return;
        S.players = S.players.map(p =>
            p.peerId === fromPeer ? { ...p, readyVote: true } : p
        );
        sendToAll({ type: 'PLAYERS_UPDATE', players: S.players });
```

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```
            const votes    = S.players.filter(p => p.readyVote).length;
            const needed = Math.ceil(S.players.length * 0.8);
            addChatMessage(`🗳️ Vote: ${votes}/${S.players.length} (cần ${needed})`,
'system');
            if (votes >= needed && S.players.length >= 2) {
                sendToAll({ type: 'START_COUNTDOWN' });
                startCountdown();
            }
            render();
            break;
        }

        // ── Chat ──────────────────────────────────────────────────────────────────────
        case 'CHAT': {
            // [BACKEND] Server would relay with timestamp + anti-spam
            addChatMessage(data.name + ': ' + data.msg, 'player', data.color);
            break;
        }

        // ── Meeting vote ──────────────────────────────────────────────────────────────
        case 'MEETING_VOTE_CAST': {
            handleMeetingVoteCast(data);
            break;
        }
        case 'MEETING_START': {
            handleMeetingStart();
            break;
        }
        case 'MEETING_CANCEL': {
            handleMeetingCancel();
            break;
        }

        // ── Coding phase ──────────────────────────────────────────────────────────────
        case 'FILE_SWITCH': {
            // [BACKEND] Server relays active-file presence to all clients
            if (!S.fileActivePlayers) S.fileActivePlayers = {};
            // Remove player from all files, then add to new file
            Object.keys(S.fileActivePlayers).forEach(f => {
                S.fileActivePlayers[f] = (S.fileActivePlayers[f] || [])
                    .filter(p => p.peerId !== data.peerId);
            });
            if (!S.fileActivePlayers[data.file]) S.fileActivePlayers[data.file] = [];
            S.fileActivePlayers[data.file].push({
                peerId: data.peerId,
                name:     data.name,
                color:    data.color
            });
            // Patch file-presence dots without full re-render
            Object.keys(FILES).forEach(fn => {
                const dotEl = document.getElementById('file-dots-' + fn);
                if (dotEl) {
                    const players = S.fileActivePlayers[fn] || [];
                    dotEl.innerHTML = players
                        .map(p => `<div title="${p.name}"
style="width:8px;height:8px;border-radius:2px;background:${p.color};flex-shrink:0;"></div>`)
```

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```
                        .join('');
                    }
                });
                break;
            }
        }
}
```

---

[ KẾT THÚC: js/network.js ]

═══════════════════════════════════════════════════════════

[LOGIC] [GAME LOGIC]
# js/game.js

═══════════════════════════════════════════════════════════

► EXPORT / KHAI BÁO:
  notify(msg, isErr)
  dbg(msg)
  addChatMessage(msg, type, color)
  broadcastChat(msg)
  copyText(text)
  fitClass(name)
  startCountdown()
  startRolePhase()
  applyRoles(vibeCoderPeerId)
  enterCodingPhase()
  startCodingTimer()
  triggerAutoRun()
  endRunResult()
  voteSkip()
  switchFile(fn)
  toggleVoice()
  cleanupRoom()

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
  network.js (sendToAll)
  tasks.js (initTasks)
  meeting-vote.js (resetMeetingVote)
  ui/render.js (render)

---

[ BẮT ĐẦU: js/game.js ]

```
/* ============================================================
   game.js — game logic & phase transitions

   Responsibility: manage phase flow, timers, role assignment.
   Reads/writes S. Calls render() after state changes.
   Calls sendToAll() to sync state to peers.

   [BACKEND NOTE] In a real architecture, most functions here
   would live on a Node.js/Go server. Clients would receive
```

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```
    phase-change events ("PHASE_CHANGED", "ROLE_ASSIGNED", etc.)
    and simply update S then render. The timers, role selection,
    and vote aggregation would all be server-authoritative.
    ============================================================ */

// ── Shared helpers ──────────────────────────────────────────────────────

function notify(msg, isErr = false) {
    S.notification = { msg, isErr };
    render();
    setTimeout(() => { S.notification = null; render(); }, 3500);
}

function dbg(msg) {
    S.debugLog.unshift('[' + new Date().toLocaleTimeString() + '] ' + msg);
    if (S.debugLog.length > 50) S.debugLog.pop();
    const el = document.getElementById('debug-log');
    if (el) el.innerHTML = S.debugLog.slice(0, 20).join('\n');
}

function addChatMessage(msg, type = 'player', color = null) {
    S.chatMessages.push({ msg, type, color, ts: Date.now() });
    render();
    setTimeout(() => {
        const el = document.getElementById('chat-scroll');
        if (el) el.scrollTop = el.scrollHeight;
    }, 30);
}

function broadcastChat(msg) {
    const full = S.myName + ': ' + msg;
    addChatMessage(full, 'player', S.myColor);
    sendToAll({ type: 'CHAT', name: S.myName, msg, color: S.myColor });
}

function copyText(text) {
    navigator.clipboard.writeText(text).then(() => notify('✅ Đã copy!'));
}

function fitClass(name) {
    if (name.length > 14) return 'fit-text fit-sm';
    if (name.length > 9)  return 'fit-text fit-md';
    return 'fit-text';
}

// ── Phase: Countdown 3-2-1 ──────────────────────────────────────────────
//   [BACKEND] Server broadcasts a COUNTDOWN_START event.
//   All clients receive it and run their own local countdown animation.

function startCountdown() {
    if (S._countdownTimer) { clearInterval(S._countdownTimer); S._countdownTimer = null; }
    S.countdown = 3;
    render();
    S._countdownTimer = setInterval(() => {
        S.countdown--;
        render();
        if (S.countdown <= 0) {
```

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```
            clearInterval(S._countdownTimer);
            S._countdownTimer = null;
            S.countdown = null;
            startRolePhase();
        }
    }, 1000);
}

// ── Phase: Role assignment ──────────────────────────────────────────────
//   [BACKEND] Server randomly picks exactly 1 vibe coder, then
//   sends each client a personalised ROLE_ASSIGNED message so
//   other clients never learn who the vibe coder is.
//   Here the host acts as a trusted peer instead.

function startRolePhase() {
    if (S.isHost) {
        const idx              = Math.floor(Math.random() * S.players.length);
        const vibeCoderPeerId = S.players[idx].peerId;
        // Tell all members their role
        sendToAll({ type: 'ROLE_ASSIGN', vibeCoderPeerId });
        // Apply to ourselves
        applyRoles(vibeCoderPeerId);
    }
    // Members call applyRoles when they receive ROLE_ASSIGN from network.js
}

function applyRoles(vibeCoderPeerId) {
    if (S._roleTimer) { clearInterval(S._roleTimer); S._roleTimer = null; }

    S.vibeCoderPeerId = vibeCoderPeerId;
    S.myRole          = (S.peerId === vibeCoderPeerId) ? 'vibecoder' : 'dev';
    S.phase           = 'role';
    S.roleTimer       = 6;
    render();

    S._roleTimer = setInterval(() => {
        S.roleTimer--;
        // Patch timer DOM directly — avoids full re-render thrash each second
        const bar = document.getElementById('role-timer-bar');
        const txt = document.getElementById('role-timer-txt');
        if (bar) bar.style.width = ((S.roleTimer / 6) * 100) + '%';
        if (txt) txt.textContent    = S.roleTimer + 's';

        if (S.roleTimer <= 0) {
            clearInterval(S._roleTimer);
            S._roleTimer = null;
            enterCodingPhase();
        }
    }, 1000);
}

// ── Phase: Coding ───────────────────────────────────────────────────────

function enterCodingPhase() {
    S.phase          = 'coding';
    S.roleTimer      = null;
    S.codingTimer    = S.roundTime * 5 || 300;
```

Chỉnh sửa bằng WPS Office

```
    S.codingTimerRunning = false;
    S.showRunResult          = false;
    if (S._codingTimerInterval) { clearInterval(S._codingTimerInterval); S._codingTimerInterval
= null; }

    // Initialise file presence tracking — put me in current file
    S.fileActivePlayers = {};
    S.fileActivePlayers[S.currentFile] = [{
        peerId: S.peerId || 'local',
        name:     S.myName,
        color:    S.myColor
    }];

    // Initialise tasks for this player's role
    initTasks();

    // Reset meeting vote state
    resetMeetingVote();

    render();
}

// ── Coding timer ──────────────────────────────────────────────────────────
//   [BACKEND] Server owns the authoritative timer and sends
//   TIMER_TICK or TIMER_EXPIRED to clients.

function startCodingTimer() {
    if (S._codingTimerInterval) return;
    S.codingTimerRunning = true;
    S._codingTimerInterval = setInterval(() => {
        S.codingTimer--;
        // Patch display without re-rendering the whole coding phase
        const timerEl = document.getElementById('coding-timer-display');
        if (timerEl) {
            const m              = Math.floor(S.codingTimer / 60);
            const sec            = S.codingTimer % 60;
            const isUrgent = S.codingTimer <= 30;
            timerEl.textContent    = m + ':' + String(sec).padStart(2, '0');
            timerEl.style.color    = isUrgent ? '#ef4444' : '#22c55e';
            const box = document.getElementById('coding-timer-box');
            if (box) box.style.borderColor = isUrgent ? '#ef444466' : '#22c55e44';
        }
        if (S.codingTimer <= 0) {
            clearInterval(S._codingTimerInterval);
            S._codingTimerInterval = null;
            S.codingTimerRunning      = false;
            S.codingTimer                 = 0;
            triggerAutoRun();
        }
    }, 1000);
}

function triggerAutoRun() {
    // [BACKEND] Server collects all code submissions and runs them server-side
    let out = `> AUTO-RUN — Tất cả file (Timer hết)\n\n`;
    Object.keys(FILES).forEach(fn => {
        const fi = FILES[fn];
```

Chỉnh sửa bằng WPS Office

```
            out += `── ${fn} (${fi.lang.toUpperCase()}) ──\n`;
            if (Math.random() < 0.25) {
                out += `❌ Error: Compilation failed\n\n`;
            } else {
                out += `✅ Compiled!\n`;
                const fake = {
                    javascript: '🎮 "Game started!"',
                    python:     '🎮 Hello from Python',
                    java:       '🎮 Hello Java Game!',
                    cpp:        '🎮 Hello C++!'
                };
                out += (fake[fi.lang] || 'Output: Game running!') + '\n\n';
            }
        });
        out += `\nProcess exited. Code 0`;

        S.terminalOutput    = out;
        S.showRunResult     = true;
        S.runResultSecs     = 15;
        S.voteSkipCounts    = 0;
        render();

        if (S._runResultTimer) clearInterval(S._runResultTimer);
        S._runResultTimer = setInterval(() => {
            S.runResultSecs--;
            const el = document.getElementById('run-result-timer');
            if (el) el.textContent = S.runResultSecs + 's';
            if (S.runResultSecs <= 0) {
                clearInterval(S._runResultTimer);
                S._runResultTimer = null;
                endRunResult();
            }
        }, 1000);
    }

    function endRunResult() {
        S.showRunResult    = false;
        S.codingTimer      = S.roundTime * 5 || 300;
        S.voteSkipCounts = 0;
        render();
        setTimeout(startCodingTimer, 100);
    }

    function voteSkip() {
        // [BACKEND] Server aggregates VOTE_SKIP messages
        S.voteSkipCounts++;
        const needed = Math.max(1, S.players.length || 1);
        const el = document.getElementById('vote-skip-count');
        if (el) el.textContent = S.voteSkipCounts + '/' + needed;
        notify('⏭ Đã vote skip (' + S.voteSkipCounts + '/' + needed + ')');
        if (S.voteSkipCounts >= needed) {
            if (S._runResultTimer) { clearInterval(S._runResultTimer); S._runResultTimer = null; }
            endRunResult();
        }
    }
```

Chỉnh sửa bằng WPS Office

```
// ── File switching ──────────────────────────────────────────────────────

function switchFile(fn) {
    const ta = document.querySelector('#code-editor-area');
    if (ta) FILES[S.currentFile].content = ta.value;

    // Update presence tracking
    Object.keys(S.fileActivePlayers).forEach(f => {
        S.fileActivePlayers[f] = (S.fileActivePlayers[f] || [])
            .filter(p => p.peerId !== (S.peerId || 'local'));
    });
    if (!S.fileActivePlayers[fn]) S.fileActivePlayers[fn] = [];
    S.fileActivePlayers[fn].push({ peerId: S.peerId || 'local', name: S.myName, color: S.myColor
 });

    sendToAll({ type: 'FILE_SWITCH', peerId: S.peerId || 'local', name: S.myName, color:
S.myColor, file: fn });
    S.currentFile = fn;
    render();
}

// ── Voice ──────────────────────────────────────────────────────────────

let _mediaStream = null;

async function toggleVoice() {
    if (S.voiceActive) {
        if (_mediaStream) { _mediaStream.getTracks().forEach(t => t.stop()); _mediaStream
= null; }
        S.voiceActive = false;
        notify('🔇 Mic đã tắt');
    } else {
        try {
            _mediaStream = await navigator.mediaDevices.getUserMedia({ audio: true,
video: false });
            S.voiceActive = true;
            notify('🎙️ Mic đang hoạt động');
        } catch (e) {
            notify('❌ Không truy cập mic: ' + e.message, true);
        }
    }
    render();
}

// ── Room lifecycle ──────────────────────────────────────────────────────

function cleanupRoom() {
    Object.values(S.connections).forEach(c => { try { c.close(); } catch (e) {} });
    S.connections = {};
    S.players = [];
    S.isHost = false;
    S.roomId = null;
    S.chatMessages = [];
    S.showChat = false;
    S.modal = null;
    S.codingTimerRunning = false;
```

Chỉnh sửa bằng WPS Office

```
    S.showRunResult          = false;
    S.codingTimer            = 300;
    if (S._codingTimerInterval) { clearInterval(S._codingTimerInterval); S._codingTimerInterval
= null; }
    if (S._runResultTimer)          { clearInterval(S._runResultTimer);          S._runResultTimer
= null; }
}
```

[ KẾT THÚC: js/game.js ]

[LOGIC] [TASK SYSTEM [MỚI]]
# js/tasks.js

► EXPORT / KHAI BÁO:
  TASKS_DEV{}
  TASKS_VIBECODER{}
  initTasks()
  completeTask(taskId)
  patchProgressBar()
  patchTaskList()
  buildTaskRow(t)

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
  game.js (gọi initTasks() tại enterCodingPhase)
  ui/taskbar.js (gọi buildTaskRow)

[ BẮT ĐẦU: js/tasks.js ]

```
/* ============================================================
   tasks.js    [LOGIC]
   Định nghĩa nhiệm vụ cho Dev và VibeCoder, logic hoàn thành,
   và tính toán tiến độ.

   ------------------------------------------------------------
   Writes to: S.tasks, S.completedTaskIds, S.taskProgress,
                    S.totalDevTasks
   Reads:         S.myRole, S.currentFile
   ============================================================ */

// ── Task definitions (keyed by filename) ──────────────────
// Dev tasks: làm tăng task progress
const TASKS_DEV = {
    'main.game.js': [
        { id: 'js-1', label: 'Viết hàm initPlayer(name)' },
        { id: 'js-2', label: 'Thêm vòng lặp gameLoop()' },
        { id: 'js-3', label: 'Khai báo biến score = 0' },
    ],
    'script.py': [
```

Chỉnh sửa bằng WPS Office

```
            { id: 'py-1', label: 'Tạo class GameState' },
            { id: 'py-2', label: 'Viết hàm check_winner()' },
        ],
        'Main.java': [
            { id: 'java-1', label: 'Implement PlayerManager class' },
            { id: 'java-2', label: 'Override phương thức toString()' },
        ],
        'game.cpp': [
            { id: 'cpp-1', label: 'Khai báo struct Player { }' },
            { id: 'cpp-2', label: 'Viết hàm update() mỗi frame' },
        ],
};

// VibeCoder tasks: sabotage — KHÔNG cộng vào progress
const TASKS_VIBECODER = {
        'main.game.js': [
            { id: 'vjs-1', label: 'Thêm while(true){} ẩn vào hàm bất kỳ' },
            { id: 'vjs-2', label: 'Đổi biến score thành s0re' },
        ],
        'script.py': [
            { id: 'vpy-1', label: 'Sửa return True thành return False' },
            { id: 'vpy-2', label: 'Thêm import thừa gây lỗi module' },
        ],
        'Main.java': [
            { id: 'vjava-1', label: 'Xóa throws Exception khỏi method' },
            { id: 'vjava-2', label: 'Comment out dòng return quan trọng' },
        ],
        'game.cpp': [
            { id: 'vcpp-1', label: 'Thêm delete ptr sau khi đã dùng' },
            { id: 'vcpp-2', label: 'Đổi <= thành < trong vòng lặp' },
        ],
};

// ── Init tasks based on role ──────────────────────────────────────────
function initTasks() {
        const source = S.myRole === 'vibecoder' ? TASKS_VIBECODER : TASKS_DEV;
        S.tasks = {};
        Object.keys(source).forEach(fn => {
            S.tasks[fn] = source[fn].map(t => ({ ...t, done: false }));
        });
        S.completedTaskIds = new Set();

        // Count total dev tasks (used for progress bar denominator)
        S.totalDevTasks = Object.values(TASKS_DEV)
            .reduce((acc, arr) => acc + arr.length, 0);
        S.taskProgress = 0;
}

// ── Complete a task ──────────────────────────────────────────────────
function completeTask(taskId) {
        // Find and mark done
        let found = false;
        Object.keys(S.tasks).forEach(fn => {
            S.tasks[fn].forEach(t => {
                if (t.id === taskId && !t.done) {
                    t.done = true;
                    found = true;
```

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```
                    S.completedTaskIds.add(taskId);
                }
            });
        });
        if (!found) return;

        // Recalculate progress (only Dev tasks count)
        if (S.myRole === 'dev') {
            const completed = [...S.completedTaskIds].filter(id => !id.startsWith('v')).length;
            S.taskProgress = S.totalDevTasks > 0
                ? Math.round((completed / S.totalDevTasks) * 100)
                : 0;
        }
        // Patch progress bar in-place to avoid full re-render
        patchProgressBar();
        // Re-render task list
        patchTaskList();
}

// ── DOM patches (avoid full re-render for perf) ──────────────────────
function patchProgressBar() {
        const bar    = document.getElementById('task-progress-fill');
        const pct    = document.getElementById('task-progress-pct');
        const note = document.getElementById('task-progress-note');
        if (bar)    bar.style.width = S.taskProgress + '%';
        if (pct)    pct.textContent = S.taskProgress + '%';
        if (note && S.taskProgress >= 100) {
            note.textContent = '🎉 Tất cả nhiệm vụ hoàn thành!';
            note.style.color = '#22c55e';
        }
}

function patchTaskList() {
        const container = document.getElementById('task-list-container');
        if (!container) return;
        const tasksForFile = (S.tasks[S.currentFile] || []);
        container.innerHTML = tasksForFile.map(t => buildTaskRow(t)).join('');
}

// Exposed for ui/taskbar.js
function buildTaskRow(t) {
        return `
        <div    style="display:flex;align-items:flex-start;gap:8px;padding:6px    0;border-bottom:1px
solid #1a1a1a;">
            <div onclick="completeTask('${t.id}')"
                style="width:16px;height:16px;border-radius:4px;flex-
shrink:0;cursor:${t.done?'default':'pointer'};margin-top:1px;
                border:1px                                                                solid
${t.done?'#22c55e':'#444'};background:${t.done?'#22c55e':'transparent'};
                display:flex;align-items:center;justify-content:center;font-size:10px;">
                ${t.done ? '✓' : ''}
            </div>
            <div style="font-size:10px;line-height:1.4;color:${t.done?'#555':'#ccc'};
                text-decoration:${t.done?'line-through':'none'};">${t.label}</div>
        </div>`;
}
```

Chỉnh sửa bằng WPS Office

```
──────────────
```

```
    S.showMeetingPrompt    = false;

    if (vote === 'yes') {
        S.meetingVoteYes++;
        S.meetingVoteActive = true; // activate banner for everyone else
        sendToAll({ type: 'MEETING_VOTE_CAST', vote: 'yes', name: S.myName, peerId:
S.peerId });
    } else {
        // Just close prompt locally — no broadcast needed for 'no' unless wanted
        sendToAll({ type: 'MEETING_VOTE_CAST', vote: 'no', name: S.myName, peerId:
S.peerId });
    }

    checkMeetingThreshold();
    render();
}

// ── Vote from another player received (via network.js) ──────────
function handleMeetingVoteCast(data) {
    if (data.peerId === S.peerId) return; // own vote already counted
    if (data.vote === 'yes') S.meetingVoteYes++;
    else                     S.meetingVoteNo++;
    S.meetingVoteActive = true;
    checkMeetingThreshold();
    render();
}

// ── Start meeting (received from network) ──────────────────────────
function handleMeetingStart() {
    resetMeetingVote();
    S.showMeeting = true;
    render();
}

// ── Cancel meeting (received from network) ──────────────────────────
function handleMeetingCancel() {
    resetMeetingVote();
    notify('❌ Cuộc họp bị hủy — không đủ phiếu');
    render();
}

// ── Threshold check (run locally on every vote) ────────────────────
function checkMeetingThreshold() {
    const total    = Math.max(1, S.players.length || 1);
    const needed = Math.ceil(total * 0.8);

    if (S.meetingVoteYes >= needed) {
        // Enough yes votes — start meeting
        sendToAll({ type: 'MEETING_START' });
        resetMeetingVote();
        S.showMeeting = true;
        return;
    }

    const remainingVoters = total - S.meetingVoteYes - S.meetingVoteNo;
    if (remainingVoters <= 0 && S.meetingVoteYes < needed) {
        // Everyone voted, not enough yes — cancel
```

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```
            sendToAll({ type: 'MEETING_CANCEL' });
            resetMeetingVote();
            notify('❌ Cuộc họp bị hủy — không đủ phiếu');
        }
}

// ── Reset all meeting-vote state ──────────────────────────────────────
function resetMeetingVote() {
        S.meetingVoteActive    = false;
        S.meetingVoteYes       = 0;
        S.meetingVoteNo        = 0;
        S.myMeetingVote        = null;
        S.showMeetingPrompt    = false;
}
```

---

[ KẾT THÚC: js/meeting-vote.js ]

---

[LOGIC] [CONTROLLER]
# js/actions.js

---

► EXPORT / KHAI BÁO:
openModal(m)
closeModal()
doCreateRoom()
doJoinRoom()
prefillJoin(id)
quickGame()
doExitRoom()
doHostStart()
doVoteStart()
doSaveSettings()
doSendChat()
doSendMeetingChat()
doRunCode()
doOpenMeeting()
doConfirmVote()
doSkipVote()

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
Được gọi bởi onclick="" trong tất cả file UI

---

[ BẮT ĐẦU: js/actions.js ]

```
/* ==============================================================
    actions.js — UI event handlers (the "controller" layer)

    Called by onclick="" in HTML templates.
    Each function reads from the DOM just enough to get user
    input, then delegates to game.js or network.js.
```

Chỉnh sửa bằng WPS Office

```
    No rendering logic lives here.
    ========================================================== */

// ── Modal helpers ──────────────────────────────────────────────────────────
function openModal(m) { S.modal = m; render(); }
function closeModal()   { S.modal = null; render(); }

// ── Lobby actions ──────────────────────────────────────────────────────────

function doCreateRoom() {
    // Read player identity from DOM (user may have typed in inputs)
    const nameEl = document.getElementById('my-name');
    if (nameEl) S.myName = nameEl.value.trim() || S.myName;

    S.roomId        = S.peerId;   // Room ID = our Peer ID so others can connect directly
    S.roomName      = document.getElementById('cr-name')?.value || 'My Room';
    S.maxPlayers    = Math.min(5, Math.max(2, parseInt(document.getElementById('cr-
max')?.value) || 5));
    S.isHost        = true;
    S.players       = [{
        peerId: S.peerId || 'local',
        name:   S.myName,
        color:  S.myColor,
        avatar: S.myAvatar,
        isHost: true,
        readyVote: false
    }];
    S.chatMessages = [];
    S.modal        = null;
    S.phase        = 'waiting';
    notify('✅ Phòng "' + S.roomName + '" đã tạo!');
    render();
}

function doJoinRoom() {
    const nameEl = document.getElementById('my-name');
    if (nameEl) S.myName = nameEl.value.trim() || S.myName;

    const hostId = (document.getElementById('join-peer-id')?.value || '').trim();
    const nick   = (document.getElementById('join-name')?.value     || S.myName).trim();

    if (!hostId) { notify('❌ Nhập Room ID của host!', true); return; }
    if (hostId === S.peerId) { notify('❌ Đó là ID phòng của chính bạn!', true); return; }

    S.myName = nick;
    S.modal  = null;
    render();

    notify('🔌 Đang kết nối tới phòng...');
    connectToPeer(hostId, conn => {
        dbg('Connected to host, sending JOIN_REQUEST');
        conn.send({ type: 'JOIN_REQUEST', name: S.myName, color: S.myColor, avatar:
S.myAvatar });
    });
}

function prefillJoin(id) {
```

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```
    openModal('join');
    setTimeout(() => {
        const el = document.getElementById('join-peer-id');
        if (el) el.value = id;
    }, 40);
}

function quickGame() {
    notify('⚡ Quick game: Tạo phòng mới để thử...');
    setTimeout(() => openModal('create'), 600);
}

// ── Waiting room actions ──────────────────────────────────────────────

function doExitRoom() {
    cleanupRoom();
    S.phase = 'lobby';
    render();
}

function doHostStart() {
    if (!S.isHost) return;
    sendToAll({ type: 'START_COUNTDOWN' });
    startCountdown();
}

function doVoteStart() {
    const host = S.players.find(p => p.isHost);
    if (host && host.peerId !== S.peerId) {
        sendTo(host.peerId, { type: 'VOTE_REQUEST' });
        notify('🗳️Đã gửi vote bắt đầu sớm');
    } else {
        doHostStart();
    }
}

function doSaveSettings() {
    const rt = parseInt(document.getElementById('s-round')?.value)     || 60;
    const mt = parseInt(document.getElementById('s-meeting')?.value) || 30;
    S.roundTime     = rt;
    S.meetingTime = mt;
    sendToAll({ type: 'SETTINGS_UPDATE', roundTime: rt, meetingTime: mt });
    notify('✅ Đã chỉnh sửa thành công!');
    closeModal();
}

// ── Chat actions ──────────────────────────────────────────────────────

function doSendChat() {
    const el = document.getElementById('chat-in');
    if (!el || !el.value.trim()) return;
    broadcastChat(el.value.trim());
    el.value = '';
}

function doSendMeetingChat() {
    const el = document.getElementById('mc-in');
```

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```
    if (!el || !el.value.trim()) return;
    S.meetingChatMessages.push({ from: S.myName, msg: el.value.trim() });
    el.value = '';
    render();
}

// ── Coding phase actions ──────────────────────────────────────────────

function doRunCode() {
    // [BACKEND] In real game: send code to server, server runs sandbox, returns result
    const fi    = FILES[S.currentFile];
    let   out         =   `>   vscode-terminal:   ${S.currentFile}   (${fi.lang.toUpperCase()})\n\n🔨
Compiling ${fi.lang}...\n`;
    if (Math.random() < 0.3) {
        out += `❌ Error: Unexpected token\n(Generic compilation failure)\n`;
    } else {
        out += `✅ Compiled successfully!\n🚀 Executing...\n`;
        const fake = {
            javascript: '🎮 "Game started!"',
            python:     '🎮 Hello from Python',
            java:       '🎮 Hello Java Game!',
            cpp:        '🎮 Hello C++!'
        };
        out += (fake[fi.lang] || 'Output: Game running!') + '\n\nProcess exited with code 0';
    }
    S.terminalOutput = out;
    S.showTerminal     = true;
    render();
}

// ── Meeting / vote actions ────────────────────────────────────────────

// ── Meeting actions ───────────────────────────────────────────────────

function doOpenMeeting() {
    // Don't open meeting immediately — show vote prompt first
    requestMeeting();
}

function doConfirmVote() {
    if (S.selectedVote) notify('✅ Vote xác nhận!');
    S.showMeeting = false;
    render();
}

function doSkipVote() {
    notify('⏭ Vote skipped');
    S.showMeeting = false;
    render();
}
```

──────────────────────────────────────────────────────────────────────────────
──────────────────────────────  [ KẾT THÚC: js/actions.js ]

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```
═══════════════════════════════════════════════════════════════════════════════
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  [UI] [RENDER CORE]
  # js/ui/render.js
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═══════════════════════════════

  ► EXPORT / KHAI BÁO:
      render()
      renderStatusBarInto(el)

  ► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
      Gọi: renderLobby(), renderWaiting(), renderRolePhase(), renderCoding()
      Được gọi bởi: game.js, network.js, actions.js, meeting-vote.js

───────────────────────────────────────────────────────────────────────────────
──────────────────────────────  [ BẮT ĐẦU: js/ui/render.js ]

/* ============================================================
   ui/render.js — main render dispatcher + shared overlays

   Responsibility: decide which phase screen to show, append
   global overlays (toast, countdown). Pure UI, no game logic.
   ============================================================ */

function render() {
    const app = document.getElementById('app');
    if (!app) return;

    let html = '';
    if      (S.phase === 'lobby')   html = renderLobby();
    else if (S.phase === 'waiting') html = renderWaiting();
    else if (S.phase === 'role')    html = renderRolePhase();
    else if (S.phase === 'coding')  html = renderCoding();

    // Global: toast notification
    if (S.notification) {
        const bg  = S.notification.isErr ? '#3b0000' : '#002200';
        const bdr = S.notification.isErr ? '#ef4444' : '#22c55e';
        const clr = S.notification.isErr ? '#fca5a5' : '#86efac';
        html += `
        <div          style="position:fixed;bottom:24px;left:50%;transform:translateX(-50%);z-
index:9999;
            background:${bg};border:1px solid ${bdr};color:${clr};
            padding:10px 20px;border-radius:10px;font-size:12px;white-space:nowrap;
            pointer-events:none;animation:modalPop 0.2s ease;">
            ${S.notification.msg}
        </div>`;
    }

    // Global: countdown overlay
    if (S.countdown !== null) {
        html += `
        <div style="position:fixed;inset:0;background:rgba(0,0,0,0.88);z-index:10000;
            display:flex;align-items:center;justify-content:center;flex-
direction:column;gap:20px;">
            <div class="countdown-anim"
```

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```
                    style="font-size:160px;font-weight:700;color:#22c55e;text-shadow:0 0 80px #22c55e60;line-height:1;">
                    ${S.countdown}
                </div>
                <div style="color:#858585;font-size:14px;letter-spacing:0.1em;">GET READY TO CODE</div>
            </div>`;
        }

        app.innerHTML = html;

        // Post-render DOM patches
        setTimeout(() => {
            renderStatusBarInto(document.getElementById('peer-status-bar'));
            const cs = document.getElementById('chat-scroll');
            if (cs) cs.scrollTop = cs.scrollHeight;
            // Start coding timer if needed (deferred so the DOM is ready)
            if (S.phase === 'coding' && !S.codingTimerRunning && !S.showRunResult && S.codingTimer > 0) {
                startCodingTimer();
            }
        }, 10);
}

function renderStatusBarInto(el) {
        if (!el) return;
        const map = {
            connecting: ['peer-wait', 'Đang kết nối...'],
            ready:      ['peer-ok',   'P2P Ready'],
            error:      ['peer-err',  'Lỗi kết nối']
        };
        const [cls, txt] = map[S.peerStatus] || ['peer-err', 'Unknown'];
        el.innerHTML = `<span class="peer-dot ${cls}"></span> <span style="font-size:10px;color:#858585">${txt}</span>`;
}
```

---

[ KẾT THÚC: js/ui/render.js ]

---

[UI] [LOBBY PHASE]
# js/ui/lobby.js

---

► EXPORT / KHAI BÁO:
renderLobby()
renderCreateModal()
renderJoinModal()

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
render.js (dispatch), actions.js (openModal, doCreateRoom, doJoinRoom)

---

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——————————————————— [ BẮT ĐẦU: js/ui/lobby.js ]

```
/* ============================================================
   ui/lobby.js — Lobby phase UI
   Pure rendering. Reads from S and constants. No logic.
   ============================================================ */

function renderLobby() {
    const modal =
        S.modal === 'create' ? renderCreateModal() :
        S.modal === 'join'   ? renderJoinModal()   : '';

    return `
    <!-- ── TOP BAR ── -->
    <div style="height:36px;background:#323233;display:flex;align-items:center;padding:0 12px;
        border-bottom:1px solid #1e1e1e;flex-shrink:0;gap:12px;">
        <div style="display:flex;gap:6px;">
            <div style="width:12px;height:12px;background:#ef4444;border-radius:50%;"></div>
            <div style="width:12px;height:12px;background:#f59e0b;border-radius:50%;"></div>
            <div style="width:12px;height:12px;background:#22c55e;border-radius:50%;"></div>
        </div>
        <span style="flex:1;text-align:center;font-size:12px;color:#cccccc;">🎮 CodeAmong — Multiplayer Coding Game</span>
        <div id="peer-status-bar" style="display:flex;align-items:center;gap:6px;"></div>
        <span style="font-size:11px;color:#858585;">${S.myName}</span>
    </div>

    <div style="display:flex;flex:1;overflow:hidden;">

        <!-- ── ACTIVITY BAR ── -->
        <div style="width:48px;background:#2d2d2e;display:flex;flex-direction:column;align-items:center;padding:8px 0;gap:16px;flex-shrink:0;">
            <div style="width:32px;height:32px;display:flex;align-items:center;justify-content:center;border-radius:8px;cursor:pointer;font-size:16px;" title="Lobby">🏠</div>
            <div style="width:32px;height:32px;display:flex;align-items:center;justify-content:center;border-radius:8px;cursor:pointer;font-size:16px;" title="Network">🌐</div>
        </div>

        <!-- ── SIDEBAR: PUBLIC ROOMS ── -->
        <div style="width:220px;background:#252526;border-right:1px solid #1e1e1e;display:flex;flex-direction:column;flex-shrink:0;">
            <div style="padding:8px 12px;font-size:10px;color:#858585;text-transform:uppercase;letter-spacing:0.1em;border-bottom:1px solid #1e1e1e;">
                Phòng Công Khai
            </div>
            <div style="flex:1;overflow:auto;">
                ${PUBLIC_ROOMS.map(r => `
                <div onclick="prefillJoin('${r.id}')"
                    style="padding:10px 12px;font-size:11px;cursor:pointer;border-bottom:1px solid #1a1a1a;
                        display:flex;justify-content:space-between;align-items:center;"
                    onmouseover="this.style.background='#2d2d2e'"
                    onmouseout="this.style.background='transparent'">
```

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```
                        <div>
                            <div style="color:#cccccc;margin-bottom:2px;">${r.name}</div>
                            <div style="color:#555;font-size:9px;">ID: ${r.id}</div>
                        </div>
                        <div style="font-size:10px;padding:2px 6px;border-radius:6px;background:#1a3a1a;color:#22c55e;">${r.cur}/${r.max}</div>
                    </div>`).join('')}
                </div>
                <!-- Debug log panel -->
                <div style="padding:8px;border-top:1px solid #1e1e1e;">
                    <div style="font-size:9px;color:#555;margin-bottom:4px;">DEBUG LOG</div>
                    <pre id="debug-log" style="font-size:8px;color:#444;max-height:80px;overflow:auto;margin:0;white-space:pre-wrap;word-break:break-all;">${S.debugLog.slice(0, 12).join('\n')}</pre>
                </div>
            </div>

            <div style="flex:1;display:flex;overflow:hidden;">

                <!-- ── CENTER: ACTION LIST ── -->
                <div style="flex:1;display:flex;flex-direction:column;align-items:center;justify-content:center;background:#1e1e1e;padding:24px;">
                    <div style="width:100%;max-width:340px;">

                        <div style="text-align:center;margin-bottom:32px;">
                            <div style="font-size:48px;margin-bottom:8px;">👾</div>
                            <div style="font-size:22px;font-weight:600;color:#007acc;margin-bottom:4px;">CodeAmong</div>
                            <div style="font-size:11px;color:#858585;">Multiplayer coding battle</div>
                        </div>

                        <!-- VS Code-style action list -->
                        <div style="font-size:9px;color:#858585;text-transform:uppercase;letter-spacing:0.12em;margin-bottom:6px;padding:0 4px;">Start</div>
                        <div style="background:#252526;border:1px solid #2d2d2e;border-radius:10px;overflow:hidden;">
                            <div class="vscode-item" onclick="openModal('create')" style="border-bottom:1px solid #1e1e1e;">
                                <span style="color:#007acc;font-size:15px;width:20px;text-align:center;">➕</span>
                                <div>
                                    <div style="font-size:13px;color:white;">Tạo phòng mới</div>
                                    <div style="font-size:9px;color:#858585;">Create a new game room</div>
                                </div>
                            </div>
                            <div class="vscode-item" onclick="openModal('join')" style="border-bottom:1px solid #1e1e1e;">
                                <span style="color:#007acc;font-size:15px;width:20px;text-align:center;">🔑</span>
                                <div>
                                    <div style="font-size:13px;color:white;">Tham gia phòng</div>
                                    <div style="font-size:9px;color:#858585;">Enter Room ID to connect</div>
```

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```
                    </div>
                </div>
                <div class="vscode-item" onclick="quickGame()">
                    <span style="color:#007acc;font-size:15px;width:20px;text-
align:center;">⚡</span>
                    <div>
                        <div style="font-size:13px;color:white;">Quick
Game</div>
                        <div style="font-size:9px;color:#858585;">Join any
available room</div>
                    </div>
                </div>
            </div>

            <!-- Peer ID display box -->
            <div style="margin-top:16px;background:#252526;border:1px solid
#2d2d2e;border-radius:10px;padding:12px;">
                <div style="font-size:9px;color:#858585;margin-bottom:6px;">📡
Room ID / Peer ID của bạn</div>
                ${S.peerId ? `
                <div style="display:flex;align-items:center;gap:8px;">
                    <code style="font-size:18px;color:#22c55e;flex:1;word-
break:break-all;line-height:1.4;letter-spacing:0.1em;text-align:center;">${S.peerId}</code>
                    <span class="copy-btn"
onclick="copyText('${S.peerId}')">Copy</span>
                </div>` : `
                <div style="font-size:10px;color:#f59e0b;display:flex;align-
items:center;gap:6px;">
                    <span class="peer-dot peer-wait"></span> Đang tạo ID
phòng...
                </div>`}
            </div>
        </div>
    </div>

    <!-- ── RIGHT PANEL: PLAYER SETUP ── -->
    <div style="width:220px;background:#252526;border-left:1px solid
#1e1e1e;padding:16px;display:flex;flex-direction:column;flex-shrink:0;">
        <div style="font-size:9px;color:#858585;text-transform:uppercase;letter-
spacing:0.1em;margin-bottom:12px;">Player Setup</div>

        <div style="margin-bottom:12px;">
            <div style="font-size:10px;color:#858585;margin-
bottom:4px;">Nickname</div>
            <input type="text" id="my-name" value="${S.myName}" maxlength="16"
                placeholder="Your name" onchange="S.myName=this.value;">
        </div>

        <div style="margin-bottom:16px;">
            <div style="font-size:10px;color:#858585;margin-bottom:8px;">Màu
sắc</div>
            <div style="display:flex;gap:8px;flex-wrap:wrap;">
                ${COLORS.map((c, i) => `
                <div
onclick="S.myColor='${c}';S.myAvatar='${AVATARS[i]}';render()"
                    style="width:28px;height:28px;border-
radius:50%;background:${c};cursor:pointer;
```

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```
                                    border:2px solid ${S.myColor === c ? 'white' : 'transparent'};transition:border 0.1s;"></div>`).join('')}
                            </div>
                        </div>

                        <div style="background:#1e1e1e;border-radius:10px;padding:10px;margin-bottom:auto;">
                            <div style="font-size:9px;color:#555;margin-bottom:6px;">Preview</div>
                            <div class="player-row" style="background:#252526;">
                                <div style="width:32px;height:32px;border-radius:8px;background:${S.myColor};display:flex;align-items:center;justify-content:center;font-size:16px;">${S.myAvatar}</div>
                                <div style="font-size:12px;color:white;">${S.myName}</div>
                            </div>
                        </div>

                        <div style="margin-top:12px;font-size:9px;color:#333;">CodeAmong · PeerJS WebRTC</div>
                    </div>
                </div>
            </div>
            ${modal}`;
}

// ── Lobby modals ──────────────────────────────────────────────────────────

function renderCreateModal() {
    return `
    <div onclick="if(event.target===this)closeModal()"
        style="position:fixed;inset:0;background:rgba(0,0,0,0.75);display:flex;align-items:center;justify-content:center;z-index:50;">
        <div class="modal-anim" style="background:#252526;border-radius:16px;padding:24px;width:380px;border:1px solid #3d3d3d;box-shadow:0 24px 60px #000a;">
            <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:20px;">
                <span style="font-size:15px;font-weight:600;color:white;">➕ Tạo Phòng Mới</span>
                <span onclick="closeModal()" style="cursor:pointer;color:#858585;font-size:18px;">✕</span>
            </div>
            <div style="display:flex;flex-direction:column;gap:12px;">
                <div>
                    <div style="font-size:10px;color:#858585;margin-bottom:4px;">Mã Phòng (Room ID)</div>
                    <input type="text" id="cr-id" value="${S.peerId || 'Đang lấy...'}"
                        style="text-align:center;font-size:18px;letter-spacing:0.12em;color:#22c55e;" readonly>
                </div>
                <div>
                    <div style="font-size:10px;color:#858585;margin-bottom:4px;">Tên phòng</div>
                    <input type="text" id="cr-name" placeholder="My Awesome Room" maxlength="30">
                </div>
                <div>
```

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```
                <div style="font-size:10px;color:#858585;margin-bottom:4px;">Số người tối đa (2–5)</div>
                <input type="number" id="cr-max" value="5" min="2" max="5">
            </div>
            <div style="background:#1a2a1a;border:1px solid #22c55e30;border-radius:8px;padding:10px;font-size:10px;color:#22c55e;">
                💡 Room ID gồm 6 chữ số. Gửi ID này để bạn bè nhập vào Join phòng.
            </div>
            <div style="display:flex;gap:10px;margin-top:4px;">
                <button class="btn btn-gray" style="flex:1;" onclick="closeModal()">Hủy</button>
                <button class="btn btn-blue" style="flex:1;" onclick="doCreateRoom()">✅ Tạo phòng</button>
            </div>
        </div>
    </div>
</div>`;
}

function renderJoinModal() {
    return `
    <div onclick="if(event.target===this)closeModal()"
        style="position:fixed;inset:0;background:rgba(0,0,0,0.75);display:flex;align-items:center;justify-content:center;z-index:50;">
        <div class="modal-anim" style="background:#252526;border-radius:16px;padding:24px;width:400px;border:1px solid #3d3d3d;box-shadow:0 24px 60px #000a;">
            <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:20px;">
                <span style="font-size:15px;font-weight:600;color:white;">🔑 Tham Gia Phòng</span>
                <span onclick="closeModal()" style="cursor:pointer;color:#858585;font-size:18px;">✕</span>
            </div>
            <div style="display:flex;flex-direction:column;gap:12px;">
                <div>
                <div style="font-size:10px;color:#858585;margin-bottom:4px;">Room ID (6 chữ số)</div>
                <input type="text" id="join-peer-id" placeholder="Ví dụ: 123456"
                    style="font-size:14px;letter-spacing:0.1em;text-align:center;">
                </div>
                <div>
                <div style="font-size:10px;color:#858585;margin-bottom:4px;">Nickname của bạn</div>
                <input type="text" id="join-name" value="${S.myName}" maxlength="16">
                </div>
                <div style="background:#1a1a2a;border:1px solid #3b82f630;border-radius:8px;padding:10px;font-size:10px;color:#93c5fd;">
                ℹ Kết nối WebRTC P2P — hoạt động qua internet nhờ TURN relay.
                </div>
                <div style="display:flex;gap:10px;margin-top:4px;">
                <button class="btn btn-gray" style="flex:1;" onclick="closeModal()">Hủy</button>
                <button class="btn btn-blue" style="flex:1;" onclick="doJoinRoom()">🚀 Kết Nối</button>
                </div>
```

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```
            </div>
        </div>
    </div>`;
}
```

[ KẾT THÚC: js/ui/lobby.js ]

[UI] [WAITING ROOM]
# js/ui/waiting.js

► EXPORT / KHAI BÁO:
renderWaiting()
renderWaitingChat()
renderConfirmExitModal()
renderSettingsModal()

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
render.js (dispatch), actions.js (doHostStart, doVoteStart, doSaveSettings)

[ BẮT ĐẦU: js/ui/waiting.js ]

```
/* ============================================================
   ui/waiting.js — Waiting room phase UI
   Pure rendering. No logic.
   ============================================================ */

function renderWaiting() {
    const voteCount    = S.players.filter(p => p.readyVote).length;
    const voteNeeded = Math.ceil(S.players.length * 0.8);

    const chatPanel = S.showChat ? renderWaitingChat() : '';
    const confirmExit = S.modal === 'confirm-exit' ? renderConfirmExitModal() : '';
    const settingsModal = S.modal === 'settings' ? renderSettingsModal() : '';

    return `
    <div style="position:fixed;inset:0;background:#1a1a1a;display:flex;align-items:center;justify-content:center;z-index:1;">
        <!-- BG grid -->
        <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(0,122,204,0.04) 1px,transparent 1px),linear-gradient(90deg,rgba(0,122,204,0.04) 1px,transparent 1px);background-size:32px 32px;pointer-events:none;"></div>

        <div style="display:flex;gap:16px;align-items:flex-start;position:relative;z-index:2;max-width:860px;width:100%;padding:20px;">

            <!-- ── MAIN TERMINAL WINDOW ── -->
            <div style="flex:1;background:#252526;border-radius:18px;border:1px solid #3d3d3d;box-shadow:0 24px 60px #000c;overflow:hidden;min-width:0;">
```

Chỉnh sửa bằng WPS Office

```
<!-- Titlebar -->
<div style="height:40px;background:#323233;display:flex;align-items:center;padding:0 14px;gap:10px;border-bottom:1px solid #1e1e1e;">
<div style="display:flex;gap:5px;">
<div style="width:12px;height:12px;background:#ef4444;border-radius:50%;"></div>
<div style="width:12px;height:12px;background:#f59e0b;border-radius:50%;"></div>
<div style="width:12px;height:12px;background:#22c55e;border-radius:50%;"></div>
</div>
<span style="font-size:11px;color:#cccccc;flex:1;text-align:center;">
🎮 ${S.roomName || 'Phòng chờ'} · Room ID: ${S.roomId} · ${S.players.length}/${S.maxPlayers}
</span>
<button onclick="S.showChat=!S.showChat;render()"
style="font-size:10px;padding:3px 10px;background:#37373d;color:#ccc;border:none;border-radius:6px;cursor:pointer;">
💬 Chat
</button>
</div>

<div style="padding:20px;">

<!-- Host: share Room ID -->
${S.isHost ? `
<div style="background:#1a2a1a;border:1px solid #22c55e30;border-radius:10px;padding:12px;margin-bottom:16px;">
<div style="font-size:9px;color:#858585;margin-bottom:6px;">📡 Chia sẻ Room ID này để người khác join:</div>
<div style="display:flex;gap:8px;align-items:center;">
<code style="font-size:18px;letter-spacing:0.1em;color:#22c55e;flex:1;word-break:break-all;">${S.peerId}</code>
<span class="copy-btn" onclick="copyText('${S.peerId}')">Copy</span>
</div>
</div>` : ''}

<!-- Player list -->
<div style="font-size:9px;color:#858585;text-transform:uppercase;letter-spacing:0.1em;margin-bottom:10px;">
Người chơi (${S.players.length}/${S.maxPlayers})
</div>
<div style="display:flex;flex-direction:column;gap:8px;margin-bottom:16px;">
${S.players.map(p => `
<div class="player-row">
<div style="width:38px;height:38px;border-radius:12px;background:${p.color};
display:flex;align-items:center;justify-content:center;font-size:18px;
flex-shrink:0;box-shadow:0 4px 12px ${p.color}40;">${p.avatar}</div>
<div style="flex:1;min-width:0;">
<div class="${fitClass(p.name)}" style="color:white;font-weight:500;">${p.name}</div>
```

Chỉnh sửa bằng WPS Office

```
                                        <div style="font-size:9px;color:#858585;">${p.isHost ? '👑
Host' : (p.readyVote ? '🗳️Vote bắt đầu' : '⏳ Chờ...')}</div>
                                        </div>
                                        <div                                style="width:7px;height:7px;border-
radius:50%;background:#22c55e;flex-shrink:0;"></div>
                                    </div>`).join('')}

                                    ${Array.from({ length: Math.max(0, S.maxPlayers - S.players.length)
 }).map(() => `
                                    <div                                        style="display:flex;align-
items:center;gap:12px;background:#1e1e1e;border:1px          dashed          #2d2d2e;border-
radius:14px;padding:10px 14px;opacity:0.4;">
                                        <div                                style="width:38px;height:38px;border-
radius:12px;background:#2d2d2e;display:flex;align-items:center;justify-
content:center;color:#444;font-size:18px;">?</div>
                                        <span        style="font-size:11px;color:#555;">Chờ        người
chơi...</span>
                                    </div>`).join('')}
                                </div>

                                <!-- Vote status (members only) -->
                                ${!S.isHost ? `
                                <div        style="text-align:center;font-size:10px;color:#858585;margin-
bottom:12px;">
                                    Vote    bắt    đầu    sớm:    ${voteCount}/${S.players.length}    (cần
≥${voteNeeded})
                                </div>` : ''}

                                <!-- Action buttons -->
                                <div style="display:flex;gap:10px;">
                                    <button          class="btn          btn-gray"          style="flex:1;"
onclick="S.modal='confirm-exit';render()">← Thoát</button>
                                    ${S.isHost ? `
                                    <button    class="btn    btn-gray"    style="padding:8px    12px;"
onclick="S.modal='settings';render()">⚙</button>
                                    <button   class="btn   btn-green"   style="flex:1;font-size:13px;font-
weight:600;" onclick="doHostStart()">▶ Bắt đầu</button>
                                    ` : `
                                    <button          class="btn          btn-amber"          style="flex:1;"
onclick="doVoteStart()">🗳️Vote bắt đầu sớm</button>
                                    `}
                                </div>
                            </div>
                        </div>

                        ${chatPanel}
                    </div>

                    ${confirmExit}
                    ${settingsModal}
                </div>`;
}

function renderWaitingChat() {
    return `
    <div        style="width:280px;background:#1e1e1e;border-radius:16px;border:1px        solid
#a855f7;display:flex;flex-direction:column;overflow:hidden;flex-shrink:0;">
```

Chỉnh sửa bằng WPS Office

```
        <div style="display:flex;justify-content:space-between;align-items:center;padding:10px 14px;border-bottom:1px solid #2d2d2e;">
            <span style="font-size:12px;color:#a855f7;font-weight:500;">💬 Chat</span>
            <div style="display:flex;gap:8px;align-items:center;">
                <button onclick="toggleVoice()"
                    style="font-size:10px;padding:3px 8px;border-radius:6px;cursor:pointer;border:none;
                    background:${S.voiceActive ? '#22c55e' : '#37373d'};color:${S.voiceActive ? '#000' : '#ccc'};"
                    class="${S.voiceActive ? 'speaking-ring' : ''}">
                    ${S.voiceActive ? '🎙 ON' : '🎙 OFF'}
                </button>
                <span onclick="S.showChat=false;render()" style="cursor:pointer;color:#858585;font-size:16px;">✕</span>
            </div>
        </div>
        <div id="chat-scroll" style="flex:1;overflow-y:auto;padding:10px;display:flex;flex-direction:column;gap:6px;max-height:300px;">
            ${S.chatMessages.map(m => `
            <div style="font-size:10px;${m.type === 'system' ? 'color:#555;text-align:center;font-style:italic;' : ''}">
                ${m.type === 'player'
                    ? `<span style="color:${m.color || '#4fc3f7'}">${m.msg.split(':')[0]}:</span><span style="color:#d4d4d4;">${m.msg.slice(m.msg.indexOf(':') + 1)}</span>`
                    : m.msg}
            </div>`).join('')}
        </div>
        <div style="display:flex;gap:6px;padding:10px;border-top:1px solid #2d2d2e;">
            <input id="chat-in" type="text" placeholder="Nhắn gì đó..." style="flex:1;font-size:11px;"
                onkeydown="if(event.key==='Enter')doSendChat()">
            <button class="btn btn-purple" style="font-size:11px;padding:6px 10px;" onclick="doSendChat()">↑</button>
        </div>
    </div>`;
}

function renderConfirmExitModal() {
    return `
    <div style="position:fixed;inset:0;background:rgba(0,0,0,0.8);display:flex;align-items:center;justify-content:center;z-index:100;">
        <div class="modal-anim" style="background:#252526;border-radius:16px;padding:24px;width:320px;border:1px solid #ef4444;text-align:center;">
            <div style="font-size:32px;margin-bottom:8px;">⚠</div>
            <div style="color:white;font-weight:500;margin-bottom:6px;">Thoát phòng?</div>
            <div style="font-size:11px;color:#858585;margin-bottom:20px;">Bạn sẽ rời khỏi phòng chờ.</div>
            <div style="display:flex;gap:10px;">
                <button class="btn btn-gray" style="flex:1;" onclick="closeModal()">Ở lại</button>
                <button class="btn btn-red" style="flex:1;" onclick="doExitRoom()">Thoát</button>
            </div>
        </div>
    </div>`;
```

Chỉnh sửa bằng WPS Office

```
}

function renderSettingsModal() {
    return `
    <div onclick="if(event.target===this)closeModal()"
        style="position:fixed;inset:0;background:rgba(0,0,0,0.75);display:flex;align-items:center;justify-content:center;z-index:50;">
        <div class="modal-anim" style="background:#252526;border-radius:16px;padding:24px;width:360px;border:1px solid #3d3d3d;">
            <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:20px;">
                <span style="font-size:15px;font-weight:600;color:white;">⚙ Cài Đặt Game</span>
                <span onclick="closeModal()" style="cursor:pointer;color:#858585;font-size:18px;">✕</span>
            </div>
            <div style="display:flex;flex-direction:column;gap:12px;">
                <div>
                    <div style="font-size:10px;color:#858585;margin-bottom:4px;">Thời gian mỗi round (giây)</div>
                    <input type="number" id="s-round" value="${S.roundTime}" min="30" max="300">
                </div>
                <div>
                    <div style="font-size:10px;color:#858585;margin-bottom:4px;">Thời gian meeting (giây)</div>
                    <input type="number" id="s-meeting" value="${S.meetingTime}" min="15" max="120">
                </div>
                <div style="display:flex;gap:10px;margin-top:4px;">
                    <button class="btn btn-gray" style="flex:1;" onclick="closeModal()">Hủy</button>
                    <button class="btn btn-green" style="flex:1;" onclick="doSaveSettings()">✅ Lưu</button>
                </div>
            </div>
        </div>
    </div>`;
}
```

[ KẾT THÚC: js/ui/waiting.js ]

[UI] [ROLE REVEAL]
# js/ui/role.js

► EXPORT / KHAI BÁO:
renderRolePhase()
renderDevScreen(playerList, timerPct)
renderVibeCoderScreen(vibeCoderPlayer, timerPct)

Chỉnh sửa bằng WPS Office

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
render.js (dispatch), game.js (applyRoles sets S.myRole)

---

[ BẮT ĐẦU: js/ui/role.js ]

```
/* ============================================================
   ui/role.js — Role reveal phase UI
   Pure rendering. Reads S.myRole to decide which screen to show.
   ============================================================ */

function renderRolePhase() {
    const isDev = S.myRole === 'dev';

    const playerList = S.players.length > 0 ? S.players
        : [{ peerId: S.peerId, name: S.myName, color: S.myColor, avatar: S.myAvatar }];

    const vibeCoderPlayer = playerList.find(p => p.peerId === S.vibeCoderPeerId)
        || { name: S.myName, color: S.myColor, avatar: S.myAvatar };

    const timerPct = ((S.roleTimer ?? 6) / 6) * 100;

    return isDev ? renderDevScreen(playerList, timerPct) : renderVibeCoderScreen(vibeCoderPlayer, timerPct);
}

function renderDevScreen(playerList, timerPct) {
    return `
    <div style="position:fixed;inset:0;display:flex;flex-direction:column;align-items:center;justify-content:center;
        background:radial-gradient(ellipse at 50% 40%, #0a2a0a 0%, #050f05 60%, #000 100%);z-index:200;overflow:hidden;">

        <!-- BG grid -->
        <div style="position:absolute;inset:0;background-image:linear-gradient(rgba(34,197,94,0.06) 1px,transparent 1px),linear-gradient(90deg,rgba(34,197,94,0.06) 1px,transparent 1px);background-size:40px 40px;pointer-events:none;"></div>
        <!-- Scan-lines -->
        <div style="position:absolute;inset:0;background:repeating-linear-gradient(0deg,transparent,transparent 2px,rgba(0,0,0,0.08) 2px,rgba(0,0,0,0.08) 4px);pointer-events:none;"></div>

        <div style="position:relative;z-index:1;display:flex;flex-direction:column;align-items:center;gap:28px;max-width:600px;width:90%;padding:20px;">

            <!-- Role badge -->
            <div style="display:flex;flex-direction:column;align-items:center;gap:10px;">
                <div style="font-size:56px;filter:drop-shadow(0 0 20px #22c55e);">💻</div>
                <div style="font-size:11px;color:#22c55e;letter-spacing:0.25em;text-transform:uppercase;">— Role Assigned —</div>
                <div style="font-size:36px;font-weight:700;color:#22c55e;text-shadow:0 0 30px #22c55e80;letter-spacing:0.05em;">
                    Bạn là Developer
                </div>
                <div style="font-size:13px;color:#86efac;opacity:0.85;">Hãy hoàn thành các nhiệm vụ trong game</div>
            </div>
```

Chỉnh sửa bằng WPS Office

```
                <!-- All teammates -->
                <div style="width:100%;">
                    <div    style="font-size:9px;color:#22c55e80;text-transform:uppercase;letter-
spacing:0.15em;text-align:center;margin-bottom:14px;">
                        Đồng đội của bạn
                    </div>
                    <div style="display:flex;flex-wrap:wrap;gap:10px;justify-content:center;">
                        ${playerList.map(p => `
                        <div                                        style="display:flex;align-
items:center;gap:8px;background:rgba(34,197,94,0.08);
                            border:1px          solid          rgba(34,197,94,0.25);border-
radius:12px;padding:8px 14px;min-width:120px;">
                            <div                          style="width:34px;height:34px;border-
radius:10px;background:${p.color};
                                display:flex;align-items:center;justify-content:center;font-
size:18px;flex-shrink:0;
                                box-shadow:0 2px 10px ${p.color}60;">${p.avatar}</div>
                            <div                style="overflow:hidden;white-space:nowrap;font-
size:${p.name.length > 10 ? '10px' : '12px'};color:white;font-weight:500;">${p.name}</div>
                        </div>`).join('')}
                    </div>
                </div>

                <!-- Timer bar -->
                <div              style="width:100%;display:flex;flex-direction:column;gap:6px;align-
items:center;">
                    <div  style="width:100%;height:4px;background:rgba(34,197,94,0.15);border-
radius:4px;overflow:hidden;">
                        <div id="role-timer-bar" style="height:100%;background:#22c55e;border
-radius:4px;transition:width 0.95s linear;width:${timerPct}%;"></div>
                    </div>
                    <div style="font-size:10px;color:#22c55e60;">
                        Vào          game          sau          <span          id="role-timer-txt"
style="color:#22c55e;">${S.roleTimer ?? 6}s</span>
                    </div>
                </div>
            </div>
        </div>`;
}

function renderVibeCoderScreen(vibeCoderPlayer, timerPct) {
    return `
    <style>
        @keyframes vignettePulse { from{opacity:0.6} to{opacity:1} }
        @keyframes          imposterReveal          {          0%{transform:scale(0.6);opacity:0}
60%{transform:scale(1.08)} 100%{transform:scale(1);opacity:1} }
    </style>
    <div                      style="position:fixed;inset:0;display:flex;flex-direction:column;align-
items:center;justify-content:center;
        background:radial-gradient(ellipse at 50% 40%, #2a0505 0%, #0f0505 60%, #000
100%);z-index:200;overflow:hidden;">

        <!-- BG grid -->
        <div                          style="position:absolute;inset:0;background-image:linear-
gradient(rgba(239,68,68,0.06) 1px,transparent 1px),linear-gradient(90deg,rgba(239,68,68,0.06)
1px,transparent 1px);background-size:40px 40px;pointer-events:none;"></div>
```

Chỉnh sửa bằng WPS Office

```
<!-- Scan-lines -->
<div style="position:absolute;inset:0;background:repeating-linear-gradient(0deg,transparent,transparent 2px,rgba(0,0,0,0.10) 2px,rgba(0,0,0,0.10) 4px);pointer-events:none;"></div>
<!-- Vignette pulse -->
<div style="position:absolute;inset:0;background:radial-gradient(ellipse at center, transparent 40%, rgba(239,68,68,0.15) 100%);animation:vignettePulse 2s ease-in-out infinite alternate;pointer-events:none;"></div>

<div style="position:relative;z-index:1;display:flex;flex-direction:column;align-items:center;gap:28px;max-width:480px;width:90%;padding:20px;">

<!-- Role badge -->
<div style="display:flex;flex-direction:column;align-items:center;gap:10px;animation:imposterReveal 0.5s cubic-bezier(.4,0,.2,1) both;">
<div style="font-size:56px;filter:drop-shadow(0 0 24px #ef444480);">🕵️
</div>
<div style="font-size:11px;color:#ef4444;letter-spacing:0.25em;text-transform:uppercase;">— Role Assigned —</div>
<div style="font-size:36px;font-weight:700;color:#ef4444;text-shadow:0 0 30px #ef444480;letter-spacing:0.05em;">
Bạn là Vibe Coder
</div>
<div style="font-size:13px;color:#fca5a5;opacity:0.85;">Phá người chơi khác trong game</div>
</div>

<!-- My player card -->
<div style="display:flex;align-items:center;gap:16px;background:rgba(239,68,68,0.1);
border:1px solid rgba(239,68,68,0.4);border-radius:18px;padding:16px 24px;
box-shadow:0 0 40px rgba(239,68,68,0.2);">
<div style="width:52px;height:52px;border-radius:14px;background:${vibeCoderPlayer.color};
display:flex;align-items:center;justify-content:center;font-size:26px;
box-shadow:0 4px 20px ${vibeCoderPlayer.color}60;">${vibeCoderPlayer.avatar}</div>
<div>
<div style="font-size:9px;color:#ef444480;letter-spacing:0.12em;text-transform:uppercase;margin-bottom:4px;">Vibe Coder</div>
<div style="font-size:16px;color:white;font-weight:600;">${vibeCoderPlayer.name}</div>
</div>
</div>

<!-- Warning -->
<div style="font-size:10px;color:#ef444460;letter-spacing:0.08em;text-align:center;line-height:1.7;">
⚠ Đừng để lộ role của bạn<br>Hành động tự nhiên như những người khác
</div>

<!-- Timer bar -->
<div style="width:100%;display:flex;flex-direction:column;gap:6px;align-items:center;">
<div style="width:100%;height:4px;background:rgba(239,68,68,0.15);border-radius:4px;overflow:hidden;">
<div id="role-timer-bar" style="height:100%;background:#ef4444;border-
```

Chỉnh sửa bằng WPS Office

```
radius:4px;transition:width 0.95s linear;width:${timerPct}%;"></div>
                    </div>
                    <div style="font-size:10px;color:#ef444460;">
                        Vào game sau <span id="role-timer-txt"
style="color:#ef4444;">${S.roleTimer ?? 6}s</span>
                    </div>
                </div>
            </div>
        </div>`;
}
```

[ KẾT THÚC: js/ui/role.js ]

[UI] [TASK SIDEBAR [MỚI]]
# js/ui/taskbar.js

► EXPORT / KHAI BÁO:
renderTaskbar()
renderProgressStrip()

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
ui/coding.js (được gọi bên trong renderCoding)
tasks.js (gọi buildTaskRow, switchFile từ onclick)

[ BẮT ĐẦU: js/ui/taskbar.js ]

```
/* ============================================================
    ui/taskbar.js    [UI]
    Right-side task panel for coding phase.
    Dev       → "Các nhiệm vụ cần hoàn thành"
    VibeCoder → "Nhiệm vụ phá hoại"
    Progress bar is rendered separately via renderProgressStrip().
    ============================================================ */

function renderTaskbar() {
    const isDev            = S.myRole === 'dev';
    const accentColor = isDev ? '#22c55e' : '#ef4444';
    const title            = isDev ? '📋 Nhiệm vụ cần hoàn thành' : '🕵️ Nhiệm vụ phá hoại';
    const tasksForFile = (S.tasks && S.tasks[S.currentFile]) || [];
    const allFiles         = S.tasks ? Object.keys(S.tasks) : [];

    // Count overall completed for this role
    const totalTasks = allFiles.reduce((a, fn) => a + (S.tasks[fn]||[]).length, 0);
    const doneTasks    = allFiles.reduce((a, fn) => a + (S.tasks[fn]||[]).filter(t=>t.done).length,
0);

    return `
    <div style="width:210px;background:#252526;border-left:1px solid #1e1e1e;
        display:flex;flex-direction:column;flex-shrink:0;overflow:hidden;">
```

Chỉnh sửa bằng WPS Office

```
        <!-- Header -->
        <div style="padding:8px 12px;font-size:10px;font-weight:600;color:${accentColor};
            border-bottom:1px solid #1e1e1e;letter-spacing:0.04em;flex-shrink:0;">
            ${title}
        </div>

        <!-- Current file label -->
        <div style="padding:6px 12px;font-size:9px;color:#555;border-bottom:1px solid
#1a1a1a;flex-shrink:0;
            background:#1e1e1e;">
            ${FILES[S.currentFile]?.icon || ''} ${S.currentFile}
        </div>

        <!-- Task list for current file -->
        <div id="task-list-container" style="flex:1;overflow-y:auto;padding:8px 12px;">
            ${tasksForFile.length === 0
                ? `<div style="font-size:10px;color:#444;text-align:center;margin-
top:20px;">Không có nhiệm vụ cho file này</div>`
                : tasksForFile.map(t => buildTaskRow(t)).join('')
            }
        </div>

        <!-- All-file summary -->
        <div style="padding:8px 12px;border-top:1px solid #1e1e1e;flex-shrink:0;">
            <div style="font-size:9px;color:#555;margin-bottom:6px;letter-
spacing:0.06em;text-transform:uppercase;">
                Tổng tiến độ cá nhân
            </div>
            <div style="display:flex;align-items:center;gap:6px;margin-bottom:8px;">
                <div style="flex:1;height:4px;background:#2d2d2e;border-
radius:2px;overflow:hidden;">
                    <div style="height:100%;background:${accentColor};border-radius:2px;

width:${totalTasks>0?Math.round(doneTasks/totalTasks*100):0}%;transition:width
0.3s;"></div>
                </div>
                <span style="font-size:9px;color:${accentColor};width:28px;text-
align:right;">
                    ${totalTasks>0?Math.round(doneTasks/totalTasks*100):0}%
                </span>
            </div>

            <!-- File switcher shortcut -->
            <div style="font-size:9px;color:#444;margin-bottom:4px;">Xem theo file:</div>
            <div style="display:flex;flex-wrap:wrap;gap:4px;">
                ${allFiles.map(fn => {
                    const tasks = S.tasks[fn] || [];
                    const done    = tasks.filter(t=>t.done).length;
                    const isActive = fn === S.currentFile;
                    return `
                    <div onclick="switchFile('${fn}')"
                        title="${fn}: ${done}/${tasks.length}"
                        style="font-size:14px;cursor:pointer;opacity:${isActive?1:0.45};

filter:${done===tasks.length&&tasks.length>0?'grayscale(0)':'grayscale(0.4)'};">
                        ${FILES[fn]?.icon || '📄'}
```

Chỉnh sửa bằng WPS Office

```
                    </div>`;
                }).join('')}
            </div>
        </div>
    </div>`;
}

// ── Thin progress strip shown at very top of coding area ──────
// (dev team progress only, VibeCoder tasks do NOT contribute)
function renderProgressStrip() {
    const pct            = S.taskProgress || 0;
    const isComplete = pct >= 100;
    const color         = isComplete ? '#22c55e'
        : pct >= 60    ? '#f59e0b'
        : pct >= 30    ? '#3b82f6'
        : '#ef4444';

    return `
    <div     style="height:6px;background:#111;flex-shrink:0;position:relative;overflow:hidden;"
title="Tiến độ nhiệm vụ Dev: ${pct}%">
        <div id="task-progress-fill"
            style="height:100%;background:${color};width:${pct}%;
            transition:width 0.5s ease,background 0.5s;
            box-shadow:0 0 6px ${color}80;">
        </div>
        <!-- Percent label + completion note -->
        <div style="position:absolute;right:8px;top:50%;transform:translateY(-50%);
            display:flex;align-items:center;gap:6px;pointer-events:none;">
            <span id="task-progress-pct"
                style="font-size:8px;color:${color};font-family:'JetBrains
Mono',monospace;line-height:1;">
                ${pct}%
            </span>
            <span id="task-progress-note"
                style="font-size:8px;color:#22c55e;font-family:'JetBrains
Mono',monospace;line-height:1;">
                ${isComplete ? '🎉 Hoàn thành 100%!' : ''}
            </span>
        </div>
    </div>`;
}
```

────────────

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:
ui/coding.js (được gọi bên trong renderCoding)
meeting-vote.js (castMeetingVote từ onclick button)

---

[ BẮT ĐẦU: js/ui/meeting-vote-ui.js ]

```
/* ============================================================
   ui/meeting-vote-ui.js   [UI]
   Hai component liên quan đến vote cuộc họp:
     1. buildMeetingPrompt()   — box nhỏ hỏi người bấm MEETING
     2. buildMeetingVoteBanner() — banner góc trên cho mọi người
   ============================================================ */

// ── Prompt nhỏ: chỉ hiện với người vừa bấm MEETING ──────────────
// Không fullscreen, không overlap nặng — compact 260×120px
function buildMeetingPrompt() {
    if (!S.showMeetingPrompt) return '';
    return `
    <div style="position:fixed;top:52px;right:100px;z-index:200;
        background:#252526;border:1px solid #f59e0b;border-radius:14px;
        padding:14px 16px;width:264px;box-shadow:0 8px 32px rgba(0,0,0,0.6);
        animation:modalPop 0.15s ease;">
        <div style="font-size:12px;color:white;font-weight:500;margin-bottom:4px;">🗣 Cuộc
họp khẩn</div>
        <div style="font-size:11px;color:#858585;margin-bottom:14px;">Bạn có muốn bắt
đầu cuộc họp?</div>
        <div style="display:flex;gap:8px;">
            <!-- ✓ Có -->
            <button onclick="castMeetingVote('yes')"

style="flex:1;height:36px;background:#22c55e;color:#000;border:none;border-radius:10px;
                font-size:16px;cursor:pointer;font-family:'JetBrains Mono',monospace;
                display:flex;align-items:center;justify-content:center;gap:6px;font-
weight:600;
                transition:background 0.1s;"
onmouseover="this.style.background='#16a34a'"
                onmouseout="this.style.background='#22c55e'">
                ✓ <span style="font-size:11px;">Có</span>
            </button>
            <!-- ✕ Không -->
            <button onclick="castMeetingVote('no')"

style="flex:1;height:36px;background:#37373d;color:#ccc;border:none;border-radius:10px;
                font-size:16px;cursor:pointer;font-family:'JetBrains Mono',monospace;
                display:flex;align-items:center;justify-content:center;gap:6px;
                transition:background 0.1s;"
onmouseover="this.style.background='#ef4444';this.style.color='#fff'"
                onmouseout="this.style.background='#37373d';this.style.color='#ccc'">
                ✕ <span style="font-size:11px;">Không</span>
            </button>
        </div>
    </div>`;
}

// ── Banner góc trên: hiện cho tất cả khi có vote đang diễn ra ──
```

Chỉnh sửa bằng WPS Office

```
// Nhỏ gọn, không che editor
function buildMeetingVoteBanner() {
    if (!S.meetingVoteActive && !S.showMeetingPrompt) return '';
    if (!S.meetingVoteActive) return ''; // chỉ show khi đã broadcast

    const total       = Math.max(1, S.players.length || 1);
    const needed      = Math.ceil(total * 0.8);
    const yes         = S.meetingVoteYes;
    const hasVoted = S.myMeetingVote !== null;

    return `
    <div style="position:fixed;top:48px;left:50%;transform:translateX(-50%);z-index:190;
        background:#1e1e1e;border:1px solid #f59e0b;border-radius:12px;
        padding:8px 14px;display:flex;align-items:center;gap:10px;
        box-shadow:0 4px 20px rgba(0,0,0,0.5);animation:modalPop 0.15s ease;
        white-space:nowrap;">
        <span style="font-size:13px;">🚨</span>
        <span    style="font-size:11px;color:#f59e0b;font-weight:500;">Tham    gia    cuộc
họp?</span>
        <span style="font-size:11px;color:#cccccc;">(${yes}/${needed} phiếu cần)</span>

        ${!hasVoted ? `
        <div style="display:flex;gap:6px;margin-left:4px;">
            <button onclick="castMeetingVote('yes')"

style="width:28px;height:28px;background:#22c55e;color:#000;border:none;border-radius:8px;
                font-size:14px;cursor:pointer;display:flex;align-items:center;justify-
content:center;font-weight:700;">
                ✓
            </button>
            <button onclick="castMeetingVote('no')"

style="width:28px;height:28px;background:#37373d;color:#ccc;border:none;border-radius:8px;
                font-size:14px;cursor:pointer;display:flex;align-items:center;justify-
content:center;">
                ✕
            </button>
        </div>` : `
        <span style="font-size:10px;color:#555;margin-left:4px;">
            Đã vote: ${S.myMeetingVote === 'yes' ? '✓ Có' : '✕ Không'}
        </span>`}
    </div>`;
}
```

---

[ KẾT THÚC: js/ui/meeting-vote-ui.js ]

---

[UI] [CODING PHASE]
# js/ui/coding.js

---

► EXPORT / KHAI BÁO:

Chỉnh sửa bằng WPS Office

```
renderCoding()
buildRunResultTerminal()
buildMeeting()
```

► ĐƯỢC GỌI BỞI / PHỤ THUỘC VÀO:

```
render.js (dispatch)
ui/taskbar.js (renderTaskbar, renderProgressStrip)
ui/meeting-vote-ui.js (buildMeetingPrompt, buildMeetingVoteBanner)
actions.js (doOpenMeeting, doConfirmVote, doSkipVote, doSendMeetingChat, voteSkip)
```

---

[ BẮT ĐẦU: js/ui/coding.js ]

```
/* ============================================================
   ui/coding.js    [UI]
   Coding phase: editor + file explorer + taskbar + overlays.
   ============================================================ */

function renderCoding() {
    const mins        = Math.floor(S.codingTimer / 60);
    const secs        = S.codingTimer % 60;
    const isUrgent = S.codingTimer <= 30;
    const timerStr = mins + ':' + String(secs).padStart(2, '0');

    const terminal          = S.showRunResult ? buildRunResultTerminal() : '';
    const meeting           = S.showMeeting     ? buildMeeting()                 : '';
    const meetingPrompt = buildMeetingPrompt();
    const meetingBanner = buildMeetingVoteBanner();

    return `
    <div      style="height:36px;background:#323233;display:flex;align-items:center;padding:0
12px;
        border-bottom:1px solid #1e1e1e;gap:8px;flex-shrink:0;position:relative;">
        <div style="display:flex;gap:5px;">
            <div                        style="width:11px;height:11px;background:#ef4444;border-
radius:50%;"></div>
            <div                        style="width:11px;height:11px;background:#f59e0b;border-
radius:50%;"></div>
            <div                        style="width:11px;height:11px;background:#22c55e;border-
radius:50%;"></div>
        </div>
        <div style="position:absolute;left:50%;transform:translateX(-50%);">
            <div                    id="coding-timer-box"                    style="display:flex;align-
items:center;gap:6px;background:#1a1a1a;
                    border:1px    solid    ${isUrgent    ?    '#ef444466'    :    '#22c55e44'};border-
radius:10px;padding:3px 12px;">
                    <span style="font-size:9px;color:#555;">⏱</span>
                    <span id="coding-timer-display"
                        style="font-size:13px;font-weight:600;color:${isUrgent    ?    '#ef4444'    :
'#22c55e'};letter-spacing:0.06em;">
                        ${timerStr}
                    </span>
            </div>
        </div>
        <span style="font-size:10px;color:#858585;margin-left:auto;">Coding Phase</span>
        <button class="btn btn-purple" style="font-size:11px;padding:5px 12px;"
            onclick="doOpenMeeting()">🗣 MEETING</button>
```

Chỉnh sửa bằng WPS Office

```
        <button class="btn btn-gray" style="font-size:11px;padding:5px 10px;"
            onclick="S.phase='lobby';render()">← Lobby</button>
    </div>

    ${renderProgressStrip()}

    <div style="display:flex;flex:1;overflow:hidden;">
        <div style="width:48px;background:#2d2d2e;display:flex;flex-direction:column;
            align-items:center;padding:8px 0;gap:4px;flex-shrink:0;">
            <div style="font-size:16px;">📁</div>
        </div>

        <div style="width:172px;background:#252526;border-right:1px solid #1e1e1e;
            flex-shrink:0;display:flex;flex-direction:column;overflow:hidden;">
            <div style="padding:8px 12px;font-size:9px;color:#858585;text-transform:uppercase;
                letter-spacing:0.1em;border-bottom:1px solid #1e1e1e;flex-shrink:0;">Explorer</div>
            <div style="flex:1;overflow-y:auto;padding:4px 0;">
                ${Object.keys(FILES).map(fn => {
                    const active = fn === S.currentFile;
                    const activePlayers = S.fileActivePlayers[fn] || [];
                    const dotRow = '<div id="file-dots-' + fn + '" style="display:flex;gap:3px;flex-wrap:nowrap;margin-top:3px;overflow:hidden;">'
                        + activePlayers.map(p => '<div title="' + p.name + '" style="width:8px;height:8px;border-radius:2px;background:' + p.color + ';flex-shrink:0;"></div>').join('')
                        + '</div>';
                    return '<div onclick="switchFile(\'' + fn + '\')"'
                        + ' style="padding:7px 10px;cursor:pointer;border-left:2px solid ' + (active ? '#007acc' : 'transparent') + ';'
                        + 'background:' + (active ? '#37373d' : 'transparent') + ';transition:background 0.1s;"'
                        + ' onmouseover="if(\'' + fn + '\'!==\'' + S.currentFile + '\')this.style.background=\'#2a2a2a\'"'
                        + ' onmouseout="if(\'' + fn + '\'!==\'' + S.currentFile + '\')this.style.background=\'transparent\'">'
                        + '<div style="display:flex;align-items:center;gap:6px;">'
                        + '<span style="font-size:13px;flex-shrink:0;">' + FILES[fn].icon + '</span>'
                        + '<span style="font-size:11px;color:' + (active ? 'white' : '#ccc') + ';white-space:nowrap;overflow:hidden;text-overflow:ellipsis;max-width:100px;">' + fn + '</span>'
                        + '</div>' + dotRow + '</div>';
                })).join('')}
            </div>
        </div>

        <div style="flex:1;display:flex;flex-direction:column;min-width:0;">
            <div style="background:#252526;display:flex;align-items:flex-end;border-bottom:1px solid #1e1e1e;padding:0 8px;flex-shrink:0;">
                <div style="display:flex;align-items:center;gap:6px;padding:6px 12px;font-size:11px;
                    color:white;background:#1e1e1e;border-bottom:2px solid #007acc;border-radius:6px 6px 0 0;">
                    ${FILES[S.currentFile].icon} ${S.currentFile}
                </div>
```

Chỉnh sửa bằng WPS Office

```
                </div>
                <div style="flex:1;padding:16px;overflow:auto;background:#1e1e1e;">
                    <textarea id="code-editor-area"

style="width:100%;height:100%;background:transparent;color:#d4d4d4;outline:none;resize:no
ne;border:none;"
                        spellcheck="false"

oninput="FILES['${S.currentFile}'].content=this.value">${FILES[S.currentFile].content}</textare
a>
                </div>
                <div style="height:24px;background:#007acc;font-size:9px;display:flex;align-
items:center;padding:0 12px;flex-shrink:0;">
                    <span style="margin-left:auto;">${S.myName} · Room ${S.roomId || 'Solo'}
 · ${S.myRole === 'vibecoder' ? '🧑‍💻 Vibe Coder' : '💻 Developer'}</span>
                </div>
            </div>

            ${renderTaskbar()}
        </div>

        ${meetingPrompt}
        ${meetingBanner}
        ${terminal}
        ${meeting}`;
}

function buildRunResultTerminal() {
    const needed = Math.max(1, S.players.length || 1);
    return `
    <div style="position:fixed;inset:0;background:rgba(0,0,0,0.82);display:flex;align-
items:center;justify-content:center;z-index:50;">
        <div class="modal-anim" style="background:#1e1e1e;border-
radius:16px;padding:20px;border:1px solid #f59e0b;width:680px;height:460px;display:flex;flex
-direction:column;">
            <div style="display:flex;justify-content:space-between;align-items:center;margin-
bottom:12px;">
                <span style="color:#22c55e;font-weight:500;">◉ AUTO-RUN — Tất cả
file</span>
                <div style="display:flex;align-items:center;gap:10px;">
                    <span style="font-size:10px;color:#555;">Tự đóng sau <span id="run-
result-timer" style="color:#f59e0b;">${S.runResultSecs}s</span></span>
                    <button onclick="voteSkip()" style="font-size:10px;padding:4px
10px;background:#37373d;color:#ccc;border:1px solid #555;border-
radius:8px;cursor:pointer;font-family:'JetBrains Mono',monospace;">
                        ⏭ Vote Skip <span id="vote-skip-
count">${S.voteSkipCounts}/${needed}</span>
                    </button>
                </div>
            </div>
            <pre style="flex:1;background:#0a0a0a;color:#22c55e;padding:16px;font-
size:11px;overflow:auto;border-radius:10px;font-family:'JetBrains
Mono',monospace;margin:0;">${S.terminalOutput}</pre>
        </div>
    </div>`;
}
```

Chỉnh sửa bằng WPS Office

```
function buildMeeting() {
    const players = S.players.length > 0 ? S.players : [
        { peerId:1, name:'Araki',     color:'#22c55e', avatar:'👾' },
        { peerId:2, name:'Player2', color:'#ef4444', avatar:'🔴' },
        { peerId:3, name:'Player3', color:'#3b82f6', avatar:'🔵' },
    ];
    const chatPanel = S.showMeetingChat ? `
    <div      style="width:260px;background:#1e1e1e;border-radius:14px;border:1px      solid
#a855f7;display:flex;flex-direction:column;overflow:hidden;flex-shrink:0;">
        <div    style="display:flex;justify-content:space-between;padding:10px    12px;border-
bottom:1px solid #2d2d2e;">
            <span style="font-size:12px;color:#a855f7;">💬 Chat</span>
            <span                                onclick="S.showMeetingChat=false;render()"
style="cursor:pointer;color:#858585;">✕</span>
        </div>
        <div            style="flex:1;overflow:auto;padding:10px;font-size:10px;display:flex;flex-
direction:column;gap:4px;max-height:280px;">
            ${S.meetingChatMessages.map(m=>'<div><span
style="color:#4fc3f7;">'+m.from+':</span>                                                <span
style="color:#d4d4d4;">'+m.msg+'</span></div>').join('')}
        </div>
        <div style="display:flex;gap:6px;padding:8px;border-top:1px solid #2d2d2e;">
            <input id="mc-in" type="text" placeholder="Nhắn..." style="flex:1;font-size:11px;"
onkeydown="if(event.key==='Enter')doSendMeetingChat()">
            <button    class="btn    btn-purple"    style="font-size:11px;padding:5px    8px;"
onclick="doSendMeetingChat()">↑</button>
        </div>
    </div>` : '';

    return `
    <div            style="position:fixed;inset:0;background:rgba(0,0,0,0.82);display:flex;align-
items:center;justify-content:center;z-index:50;">
        <div            class="modal-anim"            style="background:#252526;border-
radius:18px;padding:22px;border:2px      solid      #f59e0b;box-shadow:0      24px      60px
#000c;display:flex;flex-direction:column;max-width:860px;width:95%;">
            <div  style="display:flex;justify-content:space-between;align-items:center;margin-
bottom:18px;">
                <div style="display:flex;align-items:center;gap:10px;">
                    <span style="font-size:28px;">🚨</span>
                    <span                                style="font-size:20px;font-
weight:600;color:#f59e0b;">EMERGENCY MEETING</span>
                </div>
                <div style="display:flex;gap:8px;">
                    <button        class="btn        btn-gray"        style="font-size:11px;"
onclick="S.showMeetingChat=!S.showMeetingChat;render()">💬 Chat</button>
                    <button      class="btn      btn-red"            style="font-size:11px;"
onclick="S.showMeeting=false;render()">Đóng</button>
                </div>
            </div>
            <div style="display:flex;gap:16px;flex:1;">
                <div style="flex:1;">
                    <div  style="font-size:9px;color:#f59e0b;text-transform:uppercase;letter-
spacing:0.1em;text-align:center;margin-bottom:12px;">Chọn để vote</div>
                    <div                                style="background:#1e1e1e;border-
radius:14px;padding:20px;display:flex;flex-wrap:wrap;gap:16px;justify-
content:center;border:1px solid rgba(245,158,11,0.2);">
```

Chỉnh sửa bằng WPS Office

```
                                ${players.map(p => {
                                    const sel = S.selectedVote === (p.peerId||p.id);
                                    return '<div onclick="S.selectedVote=\'' + (p.peerId||p.id) + '\';render()'
                                        style="display:flex;flex-direction:column;align-items:center;cursor:pointer;transition:transform 0.1s;transform:' + (sel?'scale(1.12)':'scale(1)') + ';">'
                                        + '<div style="width:60px;height:60px;border-radius:14px;background:' + p.color + ';display:flex;align-items:center;justify-content:center;font-size:28px;box-shadow:0 8px 20px ' + p.color + '40;' + (sel?'outline:2px solid #22c55e;outline-offset:3px;':'') + '">' + p.avatar + '</div>'
                                        + '<div style="margin-top:8px;font-size:11px;color:white;">' + p.name + '</div>'
                                        + (sel ? '<div style="font-size:9px;color:#22c55e;">● VOTED</div>' : '')
                                        + '</div>';
                                }).join('')}
                            </div>
                            <div style="display:flex;gap:12px;justify-content:center;margin-top:16px;">
                                <button class="btn btn-green" style="padding:12px 28px;font-size:14px;" onclick="doConfirmVote()">✅ Xác nhận</button>
                                <button class="btn btn-amber" style="padding:12px 28px;font-size:14px;" onclick="doSkipVote()">⏭ Skip</button>
                            </div>
                        </div>
                        ${chatPanel}
                    </div>
                </div>
            </div>`;
}
```

---

[ KẾT THÚC: js/ui/coding.js ]

Chỉnh sửa bằng WPS Office